HOTMM: Direction-Aware Spatial–Spectral Mamba With Long-Range Memory Guidance for Hyperspectral Object Tracking

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***Abstract*—Hyperspectral object tracking (HOT) plays an increasingly important role in remote sensing surveillance, as rich spectral signatures in hyperspectral (HS) imagery enable more reliable target discrimination. However, existing HOT methods still face challenges in fully exploiting spatial–spectral–temporal interactions, including limited spatial adaptivity, insufficient spectral integration, and inadequate temporal context modeling. Although recent Mamba-based trackers offer effective sequence modeling for HOT, their predefined scanning paths with uniform aggregation tend to overlook the diverse directional structures in remote sensing scenarios. To address these challenges, this article proposes HOTMM, a novel Mamba-based framework that unifies direction-aware spatial–spectral modeling with long-range memory-guided temporal modulation to jointly capture directional spatial structures, interband spectral dependencies, and persistent target evolution. At the core of the framework is the mixture of directional Mamba experts (MDMEs) module, which leverages the mixture-of-experts (MoE) paradigm to transform such spatial–spectral cues into content-adaptive prompts. These prompts are then injected into RGB-pretrained tracking representations to enhance their discriminative capacity. Within MDME, we design a routed-and-shared expert scheme, where multiple directional spatial Mamba streams are sparsely activated and adaptively weighted based on the current spatial patterns, while bidirectional spectral scanning branches are shared to aggregate group-wise spectral features along the wavelength dimension. Furthermore, a temporal memory Mamba (TMM) mechanism is introduced to bridge deep intraframe representations with long-range interframe target evolution. By adopting a unidirectional causal scanning strategy, TMM generates memory-guided temporal modulation signals that continuously refine target descriptors under persistent appearance variations. Extensive experiments on eight public HS tracking datasets demonstrate that HOTMM achieves state-of-the-art (SOTA) tracking performance. The code will be available at https://github.com/creenciaxz/HOTMM**

***Index Terms*—Hyperspectral object tracking (HOT), Mamba, mixture-of-experts (MoEs), prompt learning, spatial–spectral–temporal modeling.**

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I. Introduction

REMOTE sensing plays a crucial role in Earth observation, enabling comprehensive analysis for environmental monitoring and resource management [1], [2], [3]. While extensive research has focused on static image analysis [4], [5], [6], [7], the growing demand for real-time surveillance has driven increasing attention toward dynamic scene understanding [8], [9]. In this context, hyperspectral object tracking (HOT) has emerged as an important research direction. Unlike conventional RGB trackers that primarily rely on spatial appearance cues and often struggle under background clutter (BC), illumination variation (IV), or visual camouflage [10], [11], hyperspectral (HS) imagery captures fine-grained spectral signatures across a wide wavelength range [12], [13]. These distinctive spectral characteristics reflect the intrinsic material properties of objects, enabling HOT to achieve more robust target localization in complex remote sensing scenarios.

In recent years, HOT has undergone a clear methodological evolution. Early studies primarily relied on handcrafted spectral features and correlation-based models to leverage material-dependent spectral responses for target discrimination [14], [15], [16]. Despite their physical interpretability, these methods often suffered from limited robustness under complex remote sensing conditions. With the rise of deep learning, HOT gradually shifted toward data-driven frameworks, where convolutional and Siamese-based trackers [17], [18], [19] significantly improved spatial–spectral representation learning, followed by transformer-based models that further enhanced global context modeling [20], [21], [22]. These advances have reinforced the consensus that effective HOT requires jointly modeling spatial structure, spectral signatures, and temporal dynamics. More recently, researchers have begun adapting large-scale RGB-pretrained foundation trackers to diverse HS domains (e.g., VIS, NIR, and RedNIR) through parameter-efficient fine-tuning [23], [24], where prompt learning has emerged as a practical strategy for addressing the scarcity of annotated HS video data.

Despite these developments, existing HOT frameworks still face several challenges in achieving effective spatial–spectral–temporal integration. Along the spatial – spectral dimension, a widely adopted strategy is to transform HS observations into false-color representations through band selection [25], [26] or band regrouping [27], [28], thereby preserving the spatial layout and introducing spectral

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cues into RGB-pretrained tracking backbones. Specifically, band selection constructs compact three-channel inputs by selecting representative bands, but may omit useful spectral information from unselected wavelengths. To alleviate this issue, band regrouping further organizes spectral bands into multiple three-channel views, thereby providing greater spectral diversity for representation learning. Nevertheless, when these false-color views are processed as separate inputs, explicit cross-group interaction can be constrained, leaving interband spectral dependencies insufficiently integrated into the tracking representation. Along the temporal dimension, most existing HS trackers remain limited to short-term contexts from adjacent frames, failing to capture long-range target evolution across extended sequences [24], [29]. In addition, existing prompt learning mechanisms predominantly rely on static prompts [21], [26], lacking the dynamic modulation required to accommodate persistent appearance variations.

Recently, advances in sequence modeling have drawn increasing attention to state space models (SSMs) [30], particularly Mamba, which supports efficient long-range dependency modeling with linear complexity. Motivated by this advantage, recent studies have begun introducing Mamba into visual tracking to better retain historical context over extended temporal horizons [31], [32]. However, existing Mamba-based tracking pipelines heavily rely on predefined scanning paths [33] and aggregate multidirectional traversals with equal or fixed fusion weights regardless of scene content [34], neglecting the directional variations inherent in remote sensing scenarios, where targets observed from aerial or satellite perspectives often exhibit arbitrary orientations and highly anisotropic spatial structures. Moreover, the inherent spectral correlations in HS data provide a meaningful basis for Mamba-based spectral dependency modeling, which remains insufficiently explored in current HOT methods.

To address these limitations, this article proposes a novel Mamba-based HOT framework named HOTMM, which performs direction-aware spatial–spectral modeling by translating directional spatial structures and interband spectral dependencies into content-adaptive spatial–spectral prompts, while encoding long-range temporal evolution as memory-guided modulation signals for robust localization under persistent appearance variations. The core component is the mixture of directional Mamba experts (MDMEs), which leverages the mixture-of-experts (MoEs) paradigm to generate such prompts through a routed-and-shared expert design. Specifically, MDME formulates multiple directional spatial Mamba streams as routed experts, where a lightweight router dynamically selects informative spatial scanning directions and assigns their fusion weights according to the current input, enabling the tracker to better accommodate the diverse structural patterns inherent in remote sensing imagery. Based on the routed spatial features, MDME further employs bidirectional spectral scanning branches as shared experts to perform group-wise spectral aggregation along the wavelength dimension, thereby incorporating interband spectral dependencies into prompt generation. The resulting spatial–spectral prompts are then injected into the tracking representations to enhance the RGB-pretrained backbone with HS-domain cues. Thereafter, reliable localization still requires explicit global interaction between the target-related template cues and the current search region. Although Mamba is well suited for structured sequence modeling, its scanning-based state propagation is not a direct way to capture such associations. Accordingly, the prompt-enhanced representations are further refined through a transformer layer, whose self-attention mechanism strengthens template–search information exchange before subsequent temporal modeling. Based on these refined representations, a temporal memory Mamba (TMM) is introduced to capture long-range target evolution along the temporal dimension. Unlike spatial or spectral modeling, TMM adopts a unidirectional scanning strategy that follows the natural chronological order of video frames, maintaining a causal state to encode the target trajectory from the first frame to the current frame. This temporal memory provides adaptive modulation signals that continuously adjust the tracking representations, enabling the tracker to reconcile deep intraframe spatial–spectral interactions with long-range interframe target evolution. The main contributions of this work are summarized as follows.

1) We propose HOTMM, a novel Mamba-based HOT framework that unifies direction-aware spatial–spectral modeling with long-range memory-guided temporal modulation to jointly capture directional spatial structures, interband spectral dependencies, and persistent target evolution.
2) An MDMEs is introduced with a routed-and-shared MoE design, which formulates multidirectional spatial scanning paths as routed experts and bidirectional spectral scanning branches as shared experts, generating content-adaptive spatial–spectral prompts to enhance RGB-pretrained tracking representations with HS-domain cues.
3) A TMM mechanism is designed to bridge deep intraframe spatial–spectral interactions with long-range interframe target evolution, thereby generating memory-guided temporal modulation signals that continuously refine tracking representations under persistent appearance variations.
4) Extensive experiments conducted on eight public HOT datasets demonstrate that HOTMM achieves state-of-the-art (SOTA) tracking performance, confirming its generalization ability and tracking stability.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed HOTMM framework, focusing on the MDME and TMM modules. Section IV reports experimental results and analysis. Section V concludes the article and outlines future work.

II. Related Work

A. HS Object Tracking

HOT entails the continuous localization of specific targets within HS video sequences, serving as a critical component in remote sensing surveillance. Early research primarily relied on correlation filters with handcrafted features. For instance, TSCFW [14] and MHT [16] incorporated tensor constraints or material-aware descriptors to enhance discrimination, yet their representation capability remained limited by shallow models. With the advent of deep learning, data-driven frameworks have dominated the field. To adapt high-dimensional HS data to pretrained RGB trackers, dimensionality reduction is widely adopted. Among these methods, band selection approaches,

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such as BABS [35] and UBSTrack [25], extract three discriminative bands based on background-aware dissimilarity or attention-based metrics. While efficient, this selection process may overlook useful responses from unselected wavelengths. Alternatively, band regrouping strategies like BAE-Net [27] and SEE-Net [36] construct multiple three-channel false-color groups according to band importance for ensemble tracking, thereby allowing trackers to benefit from diverse spectral views. Nevertheless, since these groups are processed as separate false-color inputs, the spectral dependencies among them remain insufficiently incorporated into the feature representation. Recently, prompt learning has been explored to enhance transferability. PHTrack [23] and COALT [21] introduce spectral and temporal prompts to adapt foundation models to HS tracking. Furthermore, temporal modeling approaches like SENSE [28] and SPIRIT [29] integrate motion awareness or confidence-guided updates to improve robustness. However, these methods typically rely on static prompting or short-term motion cues, failing to synergize intrinsic spectral signatures with global temporal dynamics. In contrast, by leveraging the sequence modeling capability of Mamba, our framework aggregates group-wise spectral features along the wavelength dimension to support content-adaptive spatial–spectral prompting, and further bridges deep intraframe representations with long-range interframe target evolution through memory-guided temporal modulation.

B. Visual Tracking Based on Mamba

Recent studies on Mamba exploit its linear scalability and long-range dependency modeling, effectively complementing transformer-based architectures in computer vision [37], [38]. Within the realm of RGB tracking, HA-Tracker [31] integrates Mamba into a hybrid CNN–transformer framework to capture nonlinear motion, while MCITrack [32] utilizes hidden states for video-level context propagation. However, these visible-spectrum methods often struggle in complex scenarios like camouflage, necessitating the rich material signatures inherent in HS data. Despite its immense potential, the exploration of Mamba in HOT remains in a nascent stage. Specifically, CCTrack [33] couples Mamba layers with alignment transformations to synergize spatial–spectral representations. HyMamba [34] unifies modeling via spectral state integration, synchronizing feature learning through three-directional scanning. Nevertheless, these methods typically rely on deterministic scanning paths with uniform fusion weights. In response, we leverage the MoE paradigm to adaptively select spatial scanning directions and assign content-aware fusion weights, thereby boosting accuracy for overhead remote sensing targets.

C. Mixture-of-Experts

MoEs are an effective paradigm for improving model capacity through sparse, input-dependent expert routing [39], [40], [41]. Its flexibility has led to increasing use in remote sensing. For instance, RS-MoE [42] introduces an instruction-routed MoE architecture for vision–language tasks, guiding experts to focus on distinct semantic aspects of image captioning and visual question answering. Heterogeneous MoE frameworks [43] are employed for image super-resolution, activating specialized expert groups to reconstruct diverse ground-object details. Specifically, in the field of HOT, HotMoE [44] attempts to balance efficiency and performance by regrouping spectral bands into false-color images and routing them to general-purpose experts, which can effectively exploit multiple spectral views while reducing computational cost. By contrast, our framework adopts a routed-and-shared MoE design that organizes directional spatial scanning paths as routed experts for adaptive direction selection and bidirectional spectral scanning branches as shared experts for group-wise spectral aggregation. This design helps capture informative directional spatial structures in anisotropic remote sensing targets while incorporating interband spectral dependencies into the feature representation.

III. Proposed Method

A. Overall Framework

Fig. 1 shows the overall architecture of HOTMM, which mainly consists of a HS domain adapter (HDA), a spatial–spectral–temporal interaction network, and a tracking head. The HDA first aligns heterogeneous HS and auxiliary false-color cues by learning shared cross-domain representations while preserving domain-specific characteristics. The interaction network then performs direction-aware spatial–spectral modeling with long-range memory-guided temporal modulation, after which the tracking head predicts the target state. These components are detailed below.

1) HS Domain Adapter: The inputs of HOTMM consist of one search frame and two template references. The search frame is given by an HS image $X^{HS} \in \mathbb{R}^{H\times W\times C}$ and its corresponding false-color image $X^{FRGB} \in \mathbb{R}^{H\times W\times 3}$, obtained from X^{HS} through CIE color–matching. In this design, the HS branch serves as the primary carrier of spectral information, while the false-color branch provides complementary structural cues and a lightweight interface to RGB-pretrained priors. Here, H, W denote the spatial resolution and C is the number of spectral bands. The initial template and the memory-updated template share the same modalities but are cropped at a smaller spatial scale $h \times w$, denoted as $Z^{HS}_{\text{Ini}}, Z^{HS}_{\text{Mem}} \in \mathbb{R}^{h\times w\times C}$ and $Z^{FRGB}_{\text{Ini}}, Z^{FRGB}_{\text{Mem}} \in \mathbb{R}^{h\times w\times 3}$. The Z_{Ini} provides a stable appearance anchor, whereas the Z_{Mem} is incrementally refined to encode target variation cues accumulated across the sequence.

To jointly exploit spectral information from the HS image and structural cues from the false-color image, HOTMM first projects both modalities into token representations via 3-D and 2-D patch embeddings. Taking the search frame as an example, a 3-D convolution is applied to X^{HS}, yielding $F^{HS}_X \in \mathbb{R}^{KN\times D}$, where N is the number of spatial patches, K is the number of aggregated spectral groups, and D is the embedding dimension. In parallel, a 2-D convolution is applied to X^{FRGB}, producing $F^{FRGB}_X \in \mathbb{R}^{N\times D}$. The same embeddings are applied to the initial and memory templates, resulting in $F^{HS}_{Z,\text{Ini}}, F^{HS}_{Z,\text{Mem}} \in \mathbb{R}^{Kn\times D}$ and $F^{FRGB}_{Z,\text{Ini}}, F^{FRGB}_{Z,\text{Mem}} \in \mathbb{R}^{n\times D}$, where n is the number of template spatial patches.

The HDA is then applied to these tokenized representations to capture shared structure across HS domains (e.g., VIS, NIR, and RedNIR) while retaining domain-specific characteristics. Although tokenization aligns the representation format and dimensionality, spectral discrepancies still persist across domains, motivating a private branch for each domain to model

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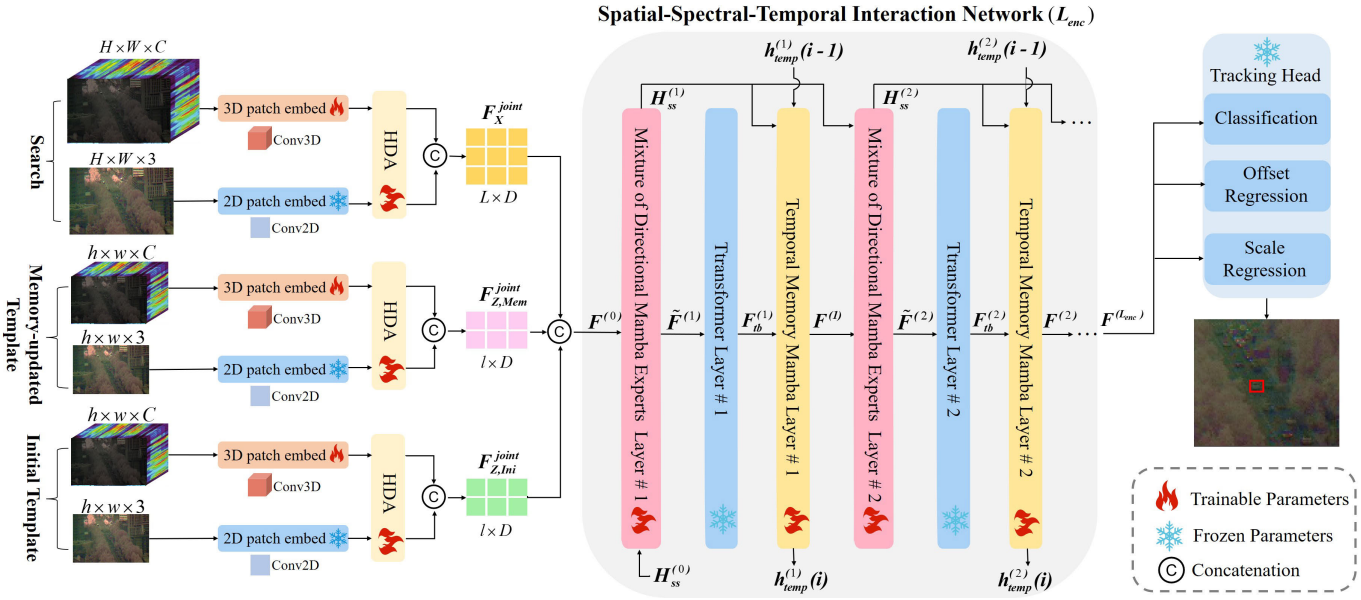


Fig. 1. Overview of our framework, comprising three components: an HDA, a spatial–spectral–temporal interaction network, and a tracking head. HDA refines tokenized HS and false-color representations through lightweight cross-domain adaptation. The interaction network stacks MDMEs, transformer, and TMM layers for direction-aware spatial–spectral modeling and long-range memory-guided temporal modulation. The final search representations are fed to the tracking head for target localization in each frame.

residual variations beyond the shared branch. Let $f \in \mathbb{R}^D$ denote a generic token and $d \in \{1, \ldots, D_{\text{dom}}\}$ its domain index. The HDA refines f as

$$\hat{f} = HDA(f) = \text{LN}\left(f + g_s(f) + g_d^{(d)}(f)\right)$$
$$g_s(f) = W_2^{(s)}\sigma\left(W_1^{(s)} f\right), \quad g_d^{(d)}(f) = W_2^{(d)}\sigma\left(W_1^{(d)} f\right) \tag{1}$$

where $g_s(\cdot)$ models domain-invariant structure, $g_d^{(d)}(\cdot)$ captures domain-specific characteristics, LN$(\cdot)$ denotes LayerNorm, and $\sigma(\cdot)$ is a ReLU activation. The projection matrices satisfy $W_1^{(\cdot)} \in \mathbb{R}^{D/\tau \times D}$ and $W_2^{(\cdot)} \in \mathbb{R}^{D \times D/\tau}$, where τ is the bottleneck ratio (set to 64). This shared–private design keeps the HDA lightweight while capturing both domain-invariant and domain-aware information.

Applying HDA to the search tokens yields $\widehat{F}_X^{HS}$ and $\widehat{F}_X^{FRGB}$, whose concatenation forms the joint search sequence $F_X^{\text{joint}} \in \mathbb{R}^{L \times D}$ with $L = N(K+1)$. Similarly, the initial and memory templates produce $F_{\text{Z,Ini}}^{\text{joint}}, F_{\text{Z,Mem}}^{\text{joint}} \in \mathbb{R}^{l \times D}$ with $l = n(K+1)$, which serve as template inputs to the subsequent spatial–spectral–temporal interaction network.

2) Spatial–Spectral–Temporal Interaction Network: Fig. 1 illustrates the core processing pipeline of HOTMM. It is implemented as a stack of L_{enc} encoder layers, each consisting of an MDME, a Transformer block, and a TMM. Given the outputs of HDA, the template and search tokens are concatenated to form the initial joint sequence $F^{(0)} \in \mathbb{R}^{(2l+L)\times D}$. For the ith frame and the ℓth encoder layer, the interaction among spatial, spectral, and temporal cues can be compactly summarized by the following three-step recursion:

$$\tilde{F}^{(\ell)},\ H_{ss}^{(\ell)} = \text{MDME}^{(\ell)}\left(F^{(\ell-1)},\ H_{ss}^{(\ell-1)}\right)$$
$$F_{\text{tb}}^{(\ell)} = \text{Transformer}^{(\ell)}\left(\tilde{F}^{(\ell)}\right)$$
$$F^{(\ell)},\ h_{\text{temp}}^{(\ell)}(i) = \text{TMM}^{(\ell)}\left(F_{\text{tb}}^{(\ell)},\ H_{ss}^{(\ell)},\ h_{\text{temp}}^{(\ell)}(i-1)\right) \tag{2}$$

where $F^{(\ell-1)}$ and $F^{(\ell)}$ denote the input and output token sets of layer ℓ, $\tilde{F}^{(\ell)}$ is the MDME-enhanced sequence, and $F_{\text{tb}}^{(\ell)}$ is the transformer output. The vector $H_{ss}^{(\ell)} = (H_{\text{spa}}^{(\ell)}, H_{\text{spec}}^{(\ell)})$ denotes the spatial–spectral memory, while $h_{\text{temp}}^{(\ell)}(i)$ is the layer-wise temporal state. For brevity, the frame index i is omitted in $F^{(\ell)}$, $F_{\text{tb}}^{(\ell)}$, $\tilde{F}^{(\ell)}$, and $H_{ss}^{(\ell)}$; these quantities are applied independently to each frame, whereas the index is retained in $h_{\text{temp}}^{(\ell)}(i)$ to emphasize temporal recursion.

Within each layer, MDME first injects spatial–spectral prompts into the joint template–search tokens and updates $H_{ss}^{(\ell)}$. The transformer block then performs multihead self-attention and feed-forward operations on the prompt-enhanced representations $\tilde{F}^{(\ell)}$, yielding $F_{\text{tb}}^{(\ell)}$ that encodes intraframe spatial relations, spectral correlations, and template–search associations. Building on this, TMM operates along the temporal axis to update $h_{\text{temp}}^{(\ell)}(i)$ from the memory-guided frame descriptor and the previous temporal state, thereby producing temporally modulated tokens $F^{(\ell)}$, which are then fed into the next-layer MDME block. Residual connections are applied around each substep to preserve low-level structure and maintain stable gradients across the deep interaction stack. In this hybrid design, MDME and TMM leverage Mamba's strength in structured sequence modeling along spatial, spectral, and temporal dimensions, whereas the transformer facilitates global template–search information exchange for reliable localization. This complementarity enables Mamba to focus on direction-aware spatial scanning, group-wise spectral aggregation, and causal temporal memory modeling, while allowing the transformer to capture global pairwise template–search interactions, forming a coherent and stable interaction pipeline. Compared with relying solely on either mechanism, this design better matches the dual requirement of HOT for structured spatial–spectral–temporal modeling and accurate template–search association.

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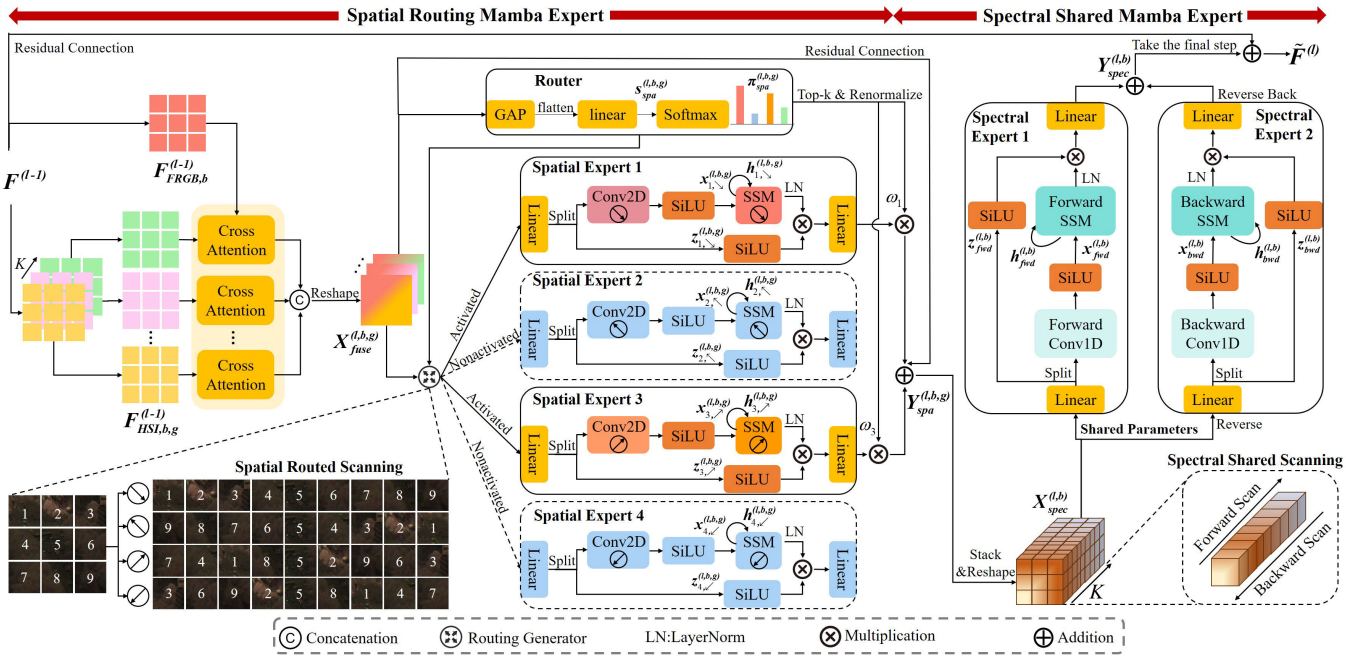


Fig. 2. Detailed architecture of the MDMEs module. For the current input, spatial experts 1 and 3 are activated by the router and assigned the corresponding routing weights w_1 and w_3, whereas spatial experts 2 and 4 remain inactive. Subsequently, spectral experts 1 and 2 capture cross-group spectral dependencies via shared branches, complementing the dynamically routed spatial features.

3) Tracking Head: The tracking head decodes the refined search tokens into an explicit bounding-box estimate. The output tokens of the search branch at the final interaction layer are reshaped into a dense feature map and fed into three parallel convolutional branches. The classification branch produces a confidence map C, where each location (x, y) indicates the probability of the target center. The scale-regression branch predicts a width–height map S, while the offset-regression branch predicts a local offset map O to compensate for quantization errors introduced by the network stride. During inference, the position with maximum confidence, $(x_c, y_c) = \arg\max_{(x,y)} C(x, y)$, is selected as the coarse target center, and the final bounding box is obtained by applying the predicted offset and scale at this position

$$\begin{aligned}(x_{box}, y_{box}) &= \left(x_c + O_x(x_c, y_c),\ \ y_c + O_y(x_c, y_c)\right)\\ (w_{box}, h_{box}) &= \left(S_w(x_c, y_c),\ \ S_h(x_c, y_c)\right).\end{aligned} \tag{3}$$

This design directly translates the enhanced representation into per-frame target localization.

B. MDMEs Module

The proposed MDME module performs direction-aware spatial modeling and group-wise spectral aggregation to generate content-adaptive spatial–spectral prompts. As illustrated in Fig. 2, MDME consists of two complementary components: 1) spatial routing Mamba experts (SRMEs), which conduct sparse and direction-selective spatial modeling, and 2) spectral shared Mamba experts (SSMEs), which model cross-group spectral dependencies through bidirectional shared scanning. Their outputs are further summarized into compact spatial–spectral memories for subsequent layers.

1) Spatial Routing Mamba Experts: The SRME aims to capture long-range spatial dependencies under multiple scanning perspectives, while sparsely activating a subset of directional experts according to the input content. At the ℓth layer, the joint token sequence $F^{(\ell-1)}$ comprises the concatenated representations of the search branch and two template branches. For a given branch $b \in \mathcal{B} = \{\mathrm{S}, \mathrm{Ini}, \mathrm{Mem}\}$, the joint tokens are partitioned into K HS group sequences $\{F^{(\ell-1)}_{\mathrm{HSI},b,g}\}_{g=1}^{K}$ and a single false-color stream $F^{(\ell-1)}_{\mathrm{FRGB},b}$. The fused prompt feature is obtained as

$$F^{(\ell)}_{\mathrm{fuse},b,g} = \mathrm{CrossAttn}\left(F^{(\ell-1)}_{\mathrm{FRGB,b}}, F^{(\ell-1)}_{\mathrm{HSI},b,g}\right) \tag{4}$$

where $\mathrm{CrossAttn}(\cdot,\cdot)$ is a cross-attention block that uses the false-color stream as query to provide a stable structural anchor, and the HSI stream as key–value to supply fine-grained material cues. Each fused sequence $F^{(\ell)}_{\mathrm{fuse},b,g}$ is reshaped into a spatial feature map $X^{(\ell,b,g)}_{\mathrm{fuse}} \in \mathbb{R}^{B\times D\times H'\times W'}$, where H' and W' denote its spatial resolution. For each $X^{(\ell,b,g)}_{\mathrm{fuse}}$, four directional scan sequences are constructed over the 2-D spatial grid. Specifically, they are generated by bidirectional row-major and column-major scanning, thereby forming a cross-scan scheme that enables each spatial location to receive contextual information from complementary traversal perspectives while maintaining a compact and symmetric set of directional scans. Letting the scan length be $L' = H'W'$, the rearrangement operator yields $X^{(\ell,b,g)}_{\mathrm{dir}} = \Phi_{4\mathrm{dir}}(X^{(\ell,b,g)}_{\mathrm{fuse}}) \in \mathbb{R}^{B\times 4\times D\times L'}$, where $\Phi_{4\mathrm{dir}}(\cdot)$ stacks the flattened sequences generated from these four scan orders, thereby enabling different experts to specialize in distinct geometric directions. A lightweight router G_{spa} predicts the gating logits for expert assignment

$$\begin{aligned}s^{(\ell,b,g)}_{\mathrm{spa}} &= G_{\mathrm{spa}}\left(X^{(\ell,b,g)}_{\mathrm{fuse}}\right)\\ &= W_{\mathrm{spa}}\mathrm{vec}\left(\mathrm{GAP}\left(X^{(\ell,b,g)}_{\mathrm{fuse}}\right)\right) \in \mathbb{R}^{B\times E_{\mathrm{spa}}}\end{aligned} \tag{5}$$

where $\mathrm{GAP}(\cdot)$ performs global average pooling over the spatial dimensions (H', W'), and $E_{\mathrm{spa}} = 4$ denotes the number of

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directional spatial experts in the proposed setting. The operator vec(·) flattens the pooled feature, while the learnable projection W_{spa} maps it to gating logits for the E_{spa} experts. The normalized gate probabilities are then given by

$$\pi_{\text{spa}}^{(\ell,b,g)} = \text{softmax}\left(s_{\text{spa}}^{(\ell,b,g)}\right). \tag{6}$$

To encourage sparsity and reduce computation, a top-k routing scheme is applied to the soft gate probabilities $\pi_{\text{spa}}^{(\ell,b,g)}$, so that only the experts corresponding to the k highest routing probabilities remain active. Since each target instance usually contains only a subset of informative directional patterns, this sparse routing mechanism serves as a spatial filter that suppresses less relevant directional responses. The active experts are then assigned masked weights $w_e^{(\ell,b,g)}$ for feature fusion and renormalized weights $\bar{w}_e^{(\ell,b,g)}$ for summarizing expert states. Each spatial expert is implemented as a direction-selective SSM associated with a specific scanning direction $\delta(e) \in \{\searrow, \nwarrow, \nearrow, \swarrow\}$. For expert $e \in \{1, 2, \ldots, E_{\text{spa}}\}$, the corresponding directional sequence is extracted from $X_{\text{dir}}^{(\ell,b,g)}$ and passed through a learnable linear projection followed by a channel-wise split, yielding $x_{e,\delta(e)}^{(\ell,b,g)}(t)$ and $z_{e,\delta(e)}^{(\ell,b,g)}(t)$, where $t = 1, \ldots, L'$ denotes the position index along the selected directional path. The former drives the state-space recurrence, whereas the latter serves as the gating input that modulates the state-space response. The hidden state is initialized using the spatial memory from the previous layer. Concretely, a lightweight decoder generates four directional prototypes given by $[h_{0,\searrow}^{(\ell)}, h_{0,\nwarrow}^{(\ell)}, h_{0,\nearrow}^{(\ell)}, h_{0,\swarrow}^{(\ell)}] = \text{Dec}_{\text{spa}}(H_{\text{spa}}^{(\ell-1)})$, and the initial state for expert e is set as $h_{e,\delta(e)}^{(\ell,b,g)}(0) = h_{0,\delta(e)}^{(\ell)}$. The resulting state-space dynamics are written as

$$\begin{aligned} h_{e,\delta(e)}^{(\ell,b,g)}(t) &= A_e h_{e,\delta(e)}^{(\ell,b,g)}(t-1) + B_e x_{e,\delta(e)}^{(\ell,b,g)}(t) \\ \tilde{y}_{e,\delta(e)}^{(\ell,b,g)}(t) &= C_e h_{e,\delta(e)}^{(\ell,b,g)}(t) + D_e x_{e,\delta(e)}^{(\ell,b,g)}(t) \\ y_{e,\delta(e)}^{(\ell,b,g)}(t) &= \text{LN}\left(\tilde{y}_{e,\delta(e)}^{(\ell,b,g)}(t)\right) \odot \text{SiLU}\left(z_{e,\delta(e)}^{(\ell,b,g)}(t)\right) \end{aligned} \tag{7}$$

where A_e, B_e, C_e, and D_e are learnable state-space parameters associated with expert e, and $\odot$ denotes element-wise multiplication. A SiLU activation on $z_{e,\delta(e)}^{(\ell,b,g)}(t)$ produces the gating signal. After directional scanning, the final hidden state $\bar{h}_{e,\delta(e)}^{(\ell,b,g)} = h_{e,\delta(e)}^{(\ell,b,g)}(L')$ is taken as the directional summary. The output sequence $\{y_{e,\delta(e)}^{(\ell,b,g)}(t)\}_{t=1}^{L'}$ is then rearranged back into a spatial feature map $Y_{e,\delta(e)}^{(\ell,b,g)} \in \mathbb{R}^{B\times D\times H'\times W'}$ and the routed spatial prompt feature is obtained via sparse fusion

$$Y_{\text{spa}}^{(\ell,b,g)} = X_{\text{fuse}}^{(\ell,b,g)} + \sum_{e=1}^{E_{\text{spa}}} w_e^{(\ell,b,g)} Y_{e,\delta(e)}^{(\ell,b,g)}. \tag{8}$$

Let $\mathcal{G} = \{(b,g) \mid b \in \{\text{S}, \text{Ini}, \text{Mem}\}, g = 1, \ldots, K\}$ denote the set of all group instances across branches, with $|\mathcal{G}|$ indicating its cardinality. For each scanning direction, the final hidden states of the selected experts are aggregated across all group instances to construct a direction-aware spatial memory

$$\bar{h}_{\delta}^{(\ell)} = \frac{1}{|\mathcal{G}|} \sum_{(b,g)\in\mathcal{G}} \sum_{e\in\mathcal{E}_\delta} \bar{w}_e^{(\ell,b,g)} \bar{h}_{e,\delta}^{(\ell,b,g)} \tag{9}$$

where $\mathcal{E}_\delta$ denotes the set of experts assigned to the specific direction $\delta \in \{\searrow, \nwarrow, \nearrow, \swarrow\}$. Finally, the direction-aware spatial memory is synthesized by concatenating the four directional summaries followed by a learnable projection:

$$H_{\text{spa}}^{(\ell)} = \Gamma_{\text{spa}}\left(\text{Concat}\left(\bar{h}_{\searrow}^{(\ell)}, \bar{h}_{\nwarrow}^{(\ell)}, \bar{h}_{\nearrow}^{(\ell)}, \bar{h}_{\swarrow}^{(\ell)}\right)\right) \tag{10}$$

where Γ_{spa} denotes a learnable linear projection. In this manner, $H_{\text{spa}}^{(\ell)}$ captures the content-adaptive preference over different spatial directions, while the sparsity induced by top-k routing ensures that only the most relevant directions are activated for each sample.

2) Spectral Shared Mamba Experts: The SSME models cross-group spectral dependencies through bidirectional shared spectral scanning, thereby aggregating spectral cues along the wavelength dimension. This shared SSM is applied across all spatial locations to consistently exploit discriminative spectral cues throughout the feature map.

For each branch b, the spatially enhanced group features $\{Y_{\text{spa}}^{(\ell,b,g)}\}_{g=1}^{K}$ are stacked along the group axis to form the spectral tensor $X_{\text{spec}}^{(\ell,b)} \in \mathbb{R}^{B\times D\times K\times H'\times W'}$. It is then reshaped into spectral sequences $X_{\text{seq}}^{(\ell,b)} \in \mathbb{R}^{(BH'W')\times K\times D}$, where each spatial location (h', w') corresponds to a length-K spectral sequence. To facilitate bidirectional modeling, we define the forward sequence as $x_{\text{fwd}}^{(\ell,b)}(k) = X_{\text{seq}}^{(\ell,b)}(k)$ and the backward sequence as $x_{\text{bwd}}^{(\ell,b)}(k) = \text{Reverse}_K(X_{\text{seq}}^{(\ell,b)})(k)$, where k indexes the spectral position. Consistent with the gated state-space formulation adopted in SRME, each spectral token $x_{\text{fwd/bwd}}^{(\ell,b)}(k)$ is linearly transformed to generate a corresponding gating term $z_{\text{fwd/bwd}}^{(\ell,b)}(k)$. The initial spectral states $h_{\text{fwd/bwd}}^{(\ell,b)}(0)$ are derived by decoding the spectral memory $H_{\text{spec}}^{(\ell-1)}$ from the previous layer. For conciseness, the bidirectional dynamics are governed by the same recurrence

$$\begin{aligned} h_{\text{dir}}^{(\ell,b)}(k) &= A_s h_{\text{dir}}^{(\ell,b)}(k-1) + B_s x_{\text{dir}}^{(\ell,b)}(k) \\ \tilde{y}_{\text{dir}}^{(\ell,b)}(k) &= C_s h_{\text{dir}}^{(\ell,b)}(k) + D_s x_{\text{dir}}^{(\ell,b)}(k) \\ y_{\text{dir}}^{(\ell,b)}(k) &= \text{LN}\left(\tilde{y}_{\text{dir}}^{(\ell,b)}(k)\right) \odot \text{SiLU}\left(z_{\text{dir}}^{(\ell,b)}(k)\right) \end{aligned} \tag{11}$$

where $\text{dir} \in \{\text{fwd}, \text{bwd}\}$ and A_s, B_s, C_s, D_s denote the shared spectral SSM parameters. The backward outputs are then reversed to enable fusion with the forward outputs at corresponding spectral positions

$$Y_{\text{spec}}^{(\ell,b)}(k) = y_{\text{fwd}}^{(\ell,b)}(k) + \text{Reverse}_K\left(y_{\text{bwd}}^{(\ell,b)}(k)\right). \tag{12}$$

We reshape $Y_{\text{spec}}^{(\ell,b)}$ into $\mathbb{R}^{B\times D\times K\times H'\times W'}$ and take the final spectral step to form a compact spectral aggregation map $\hat{Y}_{\text{spec}}^{(\ell,b)} \in \mathbb{R}^{B\times D\times H'\times W'}$. Because the bidirectional SSM sweeps across the spectral group dimension in opposite directions, $\hat{Y}_{\text{spec}}^{(\ell,b)}$ encodes global spectral context accumulated from both ends, thereby facilitating cross-group information exchange and incorporating interband dependencies into the resulting spectral features. In parallel, a layer-wise spectral memory is constructed from the final hidden states across all branches and spatial locations

$$\begin{aligned} \bar{h}_{\text{dir}}^{(\ell)} &= \frac{1}{|\mathcal{B}|H'W'} \sum_{b,h',w'} h_{\text{dir}}^{(\ell,b)}(K) \\ H_{\text{spec}}^{(\ell)} &= \Gamma_{\text{spec}}\left(\text{Concat}\left(\bar{h}_{\text{fwd}}^{(\ell)}, \bar{h}_{\text{bwd}}^{(\ell)}\right)\right) \end{aligned} \tag{13}$$

where $\mathcal{B}$ denotes the set of branches, $|\mathcal{B}|$ its cardinality, and Γ_{spec} a learnable projection. Finally, the spectral outputs $\{\hat{Y}_{\text{spec}}^{(\ell,b)}\}_{b\in\mathcal{B}}$ from all branches are projected back into the token space and concatenated into the joint token stream. The resulting tokens are then added to $F^{(\ell-1)}$ via a residual connection to form the prompt-enhanced representation $\tilde{F}^{(\ell)}$, which is subsequently fed into the transformer layer.

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C. TMM Module

The TMM module is designed to model target evolution across frames by compressing long-range dynamics into a compact temporal state, as illustrated in Fig. 3. While MDME focuses on intraframe spatial–spectral prompting, TMM operates purely along the temporal dimension: at each layer, it aggregates frame-level descriptors into a temporal state and feeds temporally modulated tokens back into the interaction pipeline. Since future frames are unavailable during inference, TMM adopts a unidirectional scan to preserve temporal causality, thereby enabling the tracker to capture target variations from the first frame to the current frame.

Let i and ℓ denote the frame and layer indices. For the ith frame, $F_{tb,i}^{(\ell)} \in \mathbb{R}^{(L+2l)\times D}$ denotes the output tokens of the ℓth transformer block, where L and l are the token numbers of the search and template streams. At the same layer, MDME also produces a spatial–spectral memory $(H_{\text{spa},i}^{(\ell)}, H_{\text{spec},i}^{(\ell)})$, which compactly encodes the directional spatial cues and bidirectional spectral correlations within the current frame.

1) Memory-Guided Token Pooling: To update the temporal state, memory-guided token pooling (MGTP) first extracts a frame-level descriptor guided by the current spatial–spectral memory. Concretely, we concatenate $(H_{\text{spa},i}^{(\ell)}, H_{\text{spec},i}^{(\ell)})$ into $H_{\text{ss},i}^{(\ell)} \in \mathbb{R}^{d_{\text{ss}}}$. A global query is then computed via a linear projection as

$$q_i^{(\ell)} = H_{\text{ss},i}^{(\ell)} W_q \tag{14}$$

where $W_q \in \mathbb{R}^{d_{\text{ss}}\times d_a}$ is a learnable projection and d_a is the attention hidden size. We use this query to perform memory-guided attention pooling over the search tokens, which carry the most up-to-date appearance of the target in frame i. Let $F_{i,S}^{(\ell)} \in \mathbb{R}^{L\times D}$ denote the subset of $F_{tb,i}^{(\ell)}$ corresponding to the search region. We construct key and value matrices as

$$K_i^{(\ell)} = F_{i,S}^{(\ell)} W_K, \quad V_i^{(\ell)} = F_{i,S}^{(\ell)} W_V \tag{15}$$

where $W_K \in \mathbb{R}^{D\times d_a}$ and $W_V \in \mathbb{R}^{D\times d_T}$ are learnable projections and d_T is the temporal descriptor dimension. The attention weights $\alpha_i^{(\ell)}$ over the L search tokens and the corresponding frame-level descriptor $x_i^{(\ell)}$ are then given by

$$\begin{aligned} \alpha_i^{(\ell)} &= \text{softmax}\left(q_i^{(\ell)} \left(K_i^{(\ell)}\right)^T \Big/ \sqrt{d_a}\right) \\ x_i^{(\ell)} &= \alpha_i^{(\ell)} V_i^{(\ell)} \in \mathbb{R}^{d_T}. \end{aligned} \tag{16}$$

In this way, MGTP leverages the spatial–spectral memory $H_{\text{ss},i}^{(\ell)}$ to adaptively weight search tokens: those whose content is more consistent with the current spatial and spectral context receive higher attention, leading to a compact descriptor $x_i^{(\ell)}$ that concentrates on the most informative target regions. For each layer ℓ, TMM maintains a temporal hidden state $h_{\text{temp}}^{(\ell)}(i) \in \mathbb{R}^{d_T}$, which summarizes the target-related dynamics from frame 1 to frame i at a fixed semantic level. The evolution of this state is modeled as a gated state-space recurrence, where a SiLU-based gating term is directly obtained via linear projection without channel-wise split to prevent unstable $x_i^{(\ell)}$ from dominating the temporal update

$$\begin{aligned} h_{\text{temp}}^{(\ell)}(i) &= A_T h_{\text{temp}}^{(\ell)}(i-1) + B_T x_i^{(\ell)} \\ \tilde{z}_i^{(\ell)} &= C_T h_{\text{temp}}^{(\ell)}(i) + D_T x_i^{(\ell)} \\ z_i^{(\ell)} &= \text{LN}\left(\tilde{z}_i^{(\ell)}\right) \odot \text{SiLU}\left(W_g x_i^{(\ell)}\right) \end{aligned} \tag{17}$$

where A_T, B_T, C_T, D_T are learnable temporal state-space parameters shared across frames, W_g is the gating weight, and $h_{\text{temp}}^{(\ell)}(0)$ is initialized to zero. Here, $h_{\text{temp}}^{(\ell)}(i)$ stores accumulated long-term dynamics, while $z_i^{(\ell)}$ provides a layer-aligned temporal readout suitable for downstream modulation.

2) Dual-Branch Temporal Prompting: To inject temporal information back into the token stream, dual-branch temporal prompting (DBTP) decouples the modulation on the search stream and the memory-updated template. Conditioned on the temporal state $z_i^{(\ell)}$, DBTP generates two compact temporal prompts

$$\left[p_{S,i}^{(\ell)}, p_{M,i}^{(\ell)}\right] = \Phi_T\left(z_i^{(\ell)}\right) \in \mathbb{R}^{2-D} \tag{18}$$

where $p_{S,i}^{(\ell)}, p_{M,i}^{(\ell)} \in \mathbb{R}^D$ correspond to the search and memory-template prompts. The mapping $\Phi_T(\cdot)$ is implemented as a two-layer MLP with a small bottleneck ratio τ: it first reduces the dimension from d_T to d_T/τ, applies a SiLU activation, and then projects to 2-D. The dual-output design avoids interference between search-specific and template-specific temporal corrections that would arise from a shared prompt.

Let $\mathcal{S}$ be the index set of search tokens, $\mathcal{T}_{\text{Mem}}$ be the memory-updated template tokens, and $\mathcal{T}_{\text{Ini}}$ be the initial template tokens. The temporally modulated tokens are given by

$$F_i^{(\ell,\text{out})}(t,:) = \begin{cases} F_{tb,i}^{(\ell)}(t,:) + p_{S,i}^{(\ell)}, & t \in \mathcal{S} \\ F_{tb,i}^{(\ell)}(t,:) + p_{M,i}^{(\ell)}, & t \in \mathcal{T}_{\text{Mem}} \\ F_{tb,i}^{(\ell)}(t,:), & t \in \mathcal{T}_{\text{Ini}}. \end{cases} \tag{19}$$

The residual design ensures that temporal prompting refines rather than overwrites the token embeddings, while keeping the initial template unchanged preserves a time-invariant appearance anchor throughout the sequence.

Finally, for each frame i and layer ℓ, the modulated tokens $F_i^{(\ell)}$ serve as input to the next MDME block, while $h_{\text{temp}}^{(\ell)}(i)$ is propagated to the next frame through (17). By combining MGTP and DBTP, the TMM bridges intraframe spatial–spectral modeling with interframe temporal aggregation, yielding a compact temporal summary that improves robustness under long-term appearance changes.

D. Training Loss

During training, the network is optimized with a unified objective that combines standard tracking supervision and an auxiliary regularization term for stabilizing the MoE routing behavior. The overall loss is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{L1}\mathcal{L}_1 + \lambda_{\text{iou}}\mathcal{L}_{\text{iou}} + \lambda_{\text{moe}}\mathcal{L}_{\text{moe}} \tag{20}$$

where $\mathcal{L}_{\text{cls}}$ is the classification loss, while $\mathcal{L}_1$ and $\mathcal{L}_{\text{iou}}$ are the bounding-box regression losses. In all experiments, $\lambda_{cls} = 1$, $\lambda_{L1} = 5$, $\lambda_{\text{iou}} = 2$, and $\lambda_{\text{moe}} = 0.1$. The first three terms supervise target localization, whereas $\mathcal{L}_{\text{moe}}$ regularizes expert selection in the MDME.

The auxiliary term $\mathcal{L}_{\text{moe}}$ addresses a known issue in sparse expert routing, where the gating network may become overly confident and repeatedly activate a small subset of experts, resulting in expert under-utilization and limited diversity. To encourage balanced expert participation while preserving sparse top-k routing, we introduce a load-balancing objective defined on the soft gating probabilities. For an SRME layer with E experts and a mini-batch of size B, let $\pi_{\text{spa}}^{(s,e)}$

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denote the softmax-normalized routing probability assigned to expert e for sample s. The regularization term is computed as

$$u_e = \frac{1}{B}\sum_{s=1}^{B}\pi_{\text{spa}}^{(s,e)}$$
$$\mathcal{L}_{\text{balance}} = \frac{1}{E}\sum_{e=1}^{E}\left(u_e - \frac{1}{E}\right)^2$$
$$\mathcal{L}_{\text{moe}} = \frac{1}{N_{\text{SRME}}}\sum_{\ell=1}^{N_{\text{SRME}}}\mathcal{L}_{\text{balance}}^{(\ell)} \tag{21}$$

where N_{SRME} denotes the number of SRME layers, u_e provides a differentiable estimate of batch-wise expert utilization, and $\mathcal{L}_{\text{balance}}$ penalizes persistent deviations from uniform expert usage. Averaging $\mathcal{L}_{\text{balance}}$ across all SRME layers stabilizes routing in a layer-consistent manner. Since the balancing loss is defined on soft gate probabilities, it remains compatible with the hard top-k sparsification applied at inference time.

IV. Experiments

A. Datasets

1) HOT2020 Dataset: HOT2020 is the first publicly available benchmark for HOT. It provides HS sequences and corresponding false-color sequences [16]. Each HS frame contains 16 spectral bands spanning approximately 470–620 nm and is captured by a snapshot mosaic HS camera. The false-color sequences are generated from the HS data using the CIE method [16]. The dataset contains 40 training sequences and 35 testing sequences, and is annotated with 11 challenging attributes, including BC, deformation (DEF), fast motion (FM), IV, in-plane rotation (IPR), low resolution (LR), motion blur (MB), occlusion (OCC), out-of-plane rotation (OPR), out-of-view (OV), and scale variation (SV).

2) HOT2023 Dataset: HOT2023 consists of three datasets [45], namely VIS2023, NIR2023, and RedNIR2023, with 16, 25, and 15 spectral bands per frame. Each dataset provides paired HS and false-color video sequences. VIS2023 includes 55 training and 46 testing sequences, NIR2023 comprises 40 training and 30 testing sequences, while RedNIR2023 contains 15 training sequences and 11 testing sequences.

3) HOT2024 Dataset: HOT2024 extends the HS tracking benchmark with three datasets [46], namely VIS2024, NIR2024, and RedNIR2024, which contain 16, 25, and 15 spectral bands per frame. Compared with HOT2023, these datasets offer a larger number of videos. Specifically, VIS2024 includes 111 training and 67 testing sequences, NIR2024 contains 70 training and 30 testing sequences, and RedNIR2024 comprises 36 training sequences and 20 testing sequences.

4) IMEC25 Dataset: IMEC25 consists of 135 annotated HS video sequences recorded at 10 frames per second [47]. It is split into 55 training sequences and 80 testing sequences. Each HS frame contains 25 spectral bands. In addition, IMEC25 is annotated with 11 challenging attributes, which are consistent with those defined in the HOT2020 benchmark. Representative tracking examples selected from these HOT benchmarks are presented in Fig. 4.

B. Evaluation Metrics

Tracking performance is evaluated under the one-pass evaluation (OPE) protocol [48] using success and precision plots. The success plot measures the proportion of frames in which the overlap between the predicted bounding box and the ground truth exceeds a given threshold, while the precision plot reports the percentage of frames whose center location error falls within a specified distance threshold. For quantitative comparison, the area under the success curve (AUC) and the distance precision at a 20-pixel threshold (DP@20) are adopted as the primary evaluation metrics. Higher AUC and DP@20 values indicate better tracking performance.

C. Implementation Details

All training experiments are conducted on a server equipped with an Intel Xeon Gold 6271C CPU and two NVIDIA RTX 4090 GPUs, each with 24 GB of memory. The inference speed is evaluated on a single RTX 4090 GPU. HOTMM is implemented in PyTorch and initialized from the large-scale RGB-pretrained foundation model in [49]. During training, only the HDA, MDME, and TMM modules, together with the 3-D convolutional patch embedding layers, are optimized with a learning rate of 2×10^{-4}, while the HiViT-based [50] backbone and the tracking head remain frozen. The model is trained for 20 epochs, with the learning rate decayed by a factor of 10 after the 16th epoch. The AdamW optimizer is used with a weight decay of 10^{-4}, and the training batch size is set to 8. The input search regions are resized to 256×256 pixels, while both the initial and memory-updated templates are resized to 128×128 pixels.

D. Comparison on HOT2020 Dataset

In this experiment, we compare HOTMM with SOTA RGB and HS trackers on the HOT2020 dataset to validate its effectiveness. The corresponding overall success and precision plots are shown in Figs. 5(l) and 6(l), respectively.

1) Comparison With RGB Trackers: HOTMM is compared with 11 SOTA RGB trackers, including correlation filter-based, transformer-based, and Mamba-based methods. In this comparison, HOTMM is evaluated on HS sequences, whereas the RGB trackers are tested on the corresponding false-color sequences. The same protocol is consistently adopted in later evaluations. It should be noted that false-color sequences provide RGB-compatible three-channel representations, while HS sequences contain richer spectral information. Therefore, this comparison mainly reflects the benefit of leveraging richer spectral cues and HS-oriented modeling. As reported in Table I, HOTMM achieves the best performance among all compared RGB trackers. In particular, compared with the strongest RGB baseline, SUTrack, HOTMM improves AUC by 6.9% and DP@20 by 3.0%. These observations suggest that richer spectral cues, when properly modeled, can contribute to more reliable target discrimination in complex tracking scenarios.

2) Comparison With HS Trackers: We further compare HOTMM with 12 SOTA HS trackers. As shown in Table I, HOTMM achieves the best overall performance among all competing HS methods. Compared with the handcrafted tracker MHT, HOTMM improves AUC and DP@20 by 16.2% and 8.5%. In addition, BAE-Net, SiamBAG, SENSE, SPIRIT,

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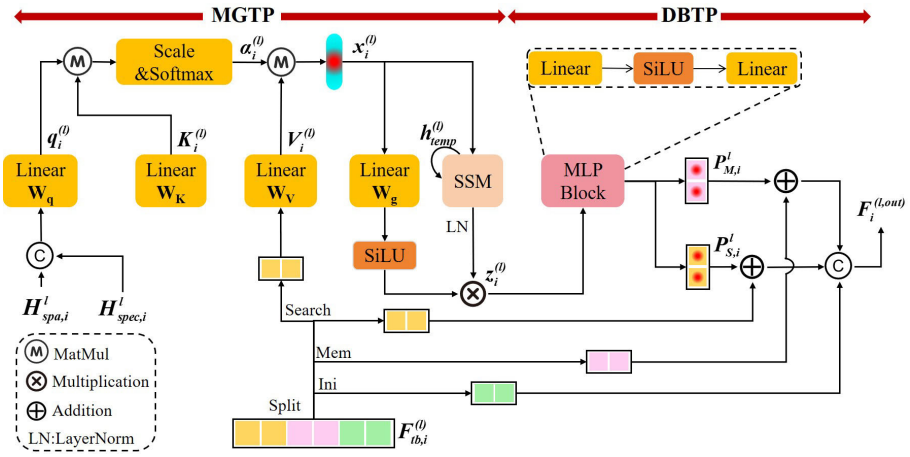


Fig. 3. Detailed architecture of the TMM module. MGTP extracts a memory-guided frame-level descriptor, after which a gated state-space recurrence updates the temporal state. DBTP then generates decoupled temporal prompts for the search stream and the memory-updated template.

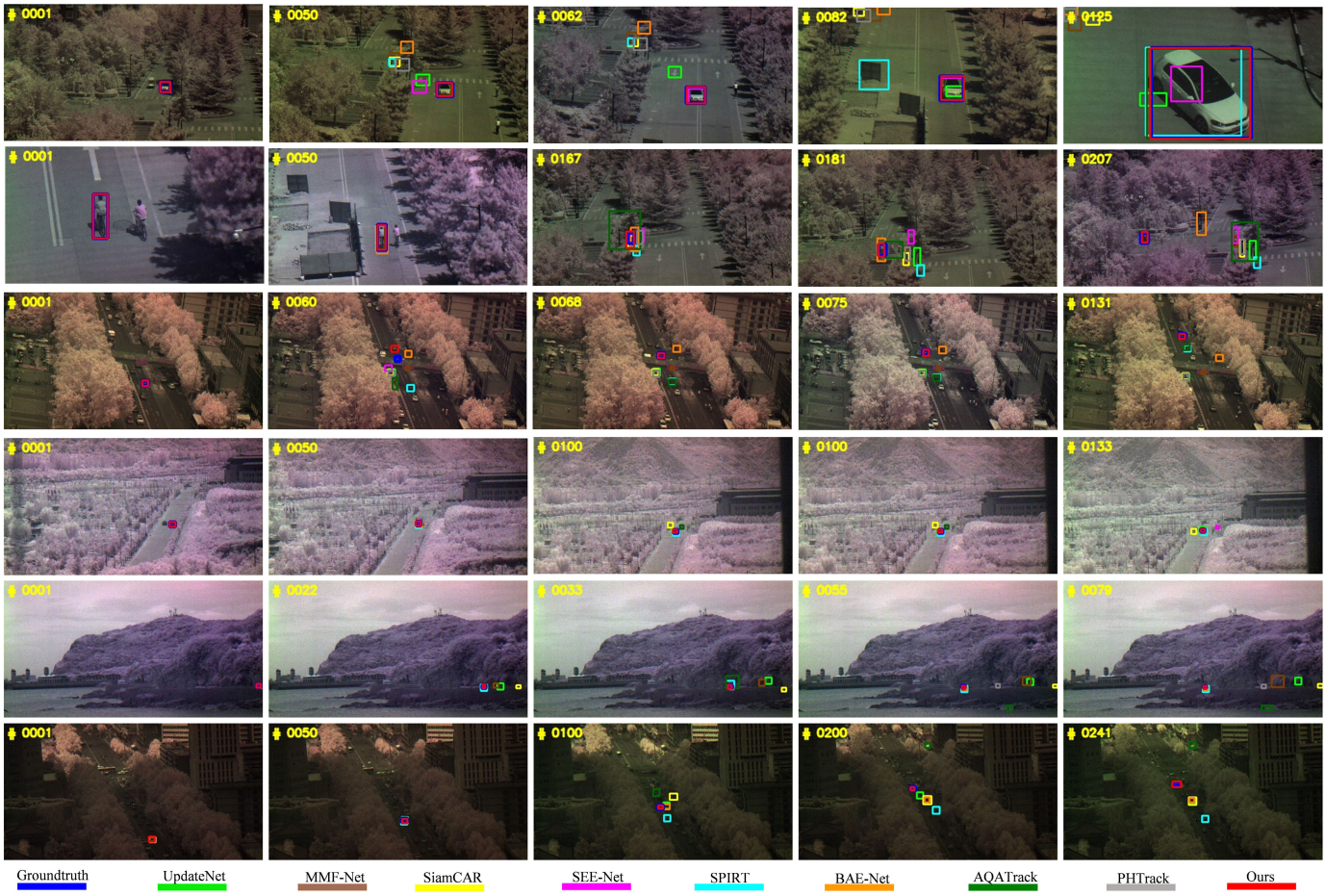


Fig. 4. Qualitative evaluation of various trackers. The sequences displayed from top to bottom correspond to car1, doublebicycle, doublecar3, doublecar, motorcycle1, and doublecar7 from the HOT benchmark.

and HotMoE transform HS data into multiple three-channel representations for tracking, which can provide diverse spectral views but may weaken coherent interband dependencies. Among them, HotMoE achieves the best AUC of 70.4%, while SENSE attains the highest DP@20 of 95.0%. By contrast, HOTMM surpasses them by 4.5% in AUC and 1.5% in DP@20. HOTMM also outperforms MMF-Net, which adopts spectral unmixing for multisource fusion, by 6.0% in AUC and 3.4% in DP@20. Compared with the short-term temporal tracker DaSSP-Net and the prompt-based tracker PHTrack, HOTMM improves AUC by 6.9% and 8.9%, and DP@20 by 4.9% and 4.7%. In addition, HOTMM significantly outperforms the Mamba-based tracker CCTrack, with gains of 6.6% in AUC and 4.2% in DP@20. Finally, HOTMM surpasses SpectralTrack, which achieves the best competing AUC, by 2.4%, and exceeds SSTtrack, which attains the

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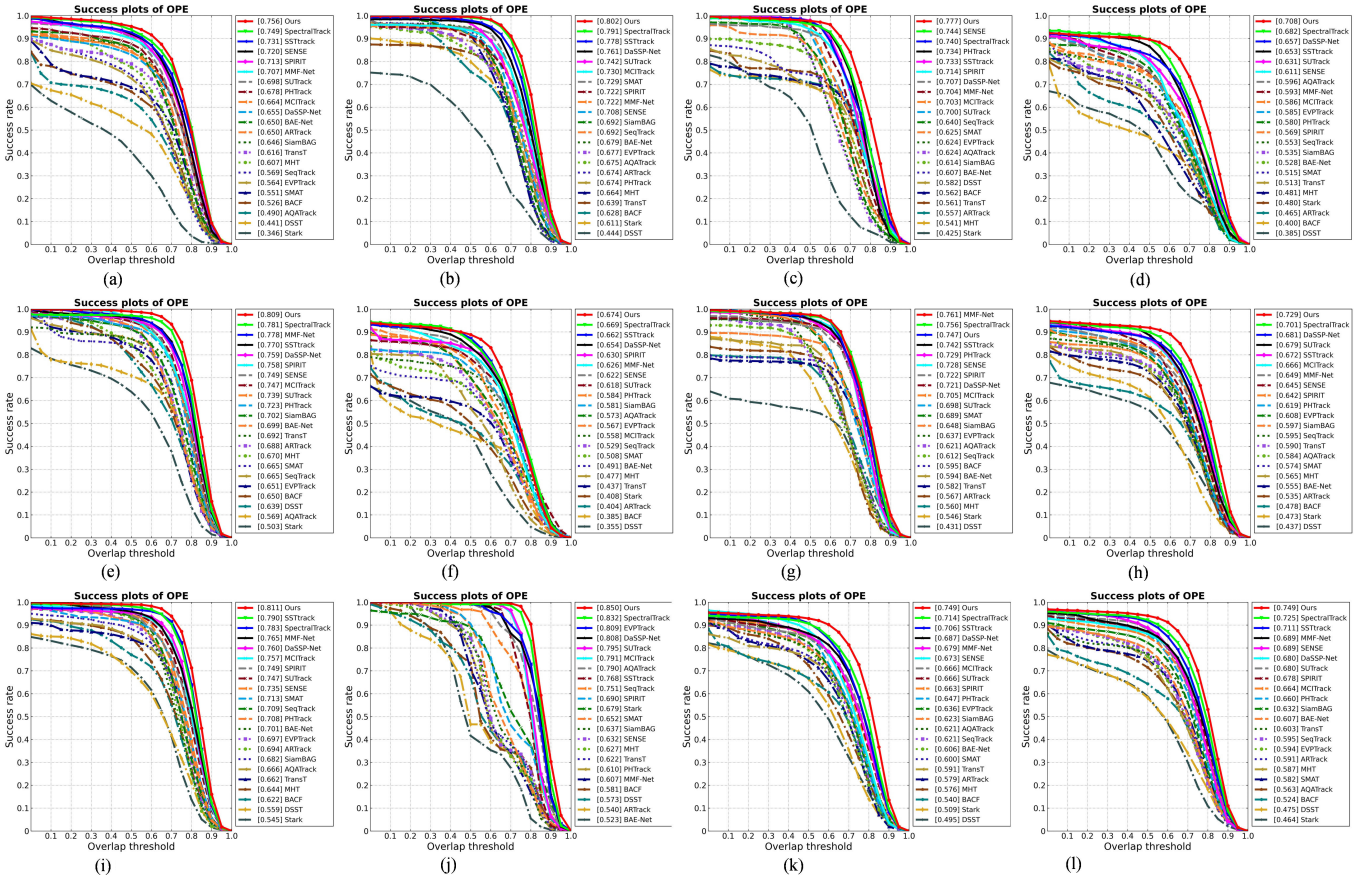


Fig. 5. Attribute-wise and overall success plots of RGB and HS trackers on the HOT2020 dataset. (a)–(k) Results under the attributes BC, DEF, FM, IV, IPR, LR, MB, OCC, OPR, OV, and SV. (l) Overall success plot.

TABLE I

COMPARISON WITH SOTA TRACKERS ON THE HOT2020 DATASET. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN RED AND BLUE

RGB Trackers	DSST [51]	BACF [52]	TransT [53]	ARTrack [54]	SeqTrack [55]	AQATrack [56]	Stark [57]	MCITrack [32]	SMAT [58]	EVPTrack [59]	SUTrack [60]
AUC	0.475	0.524	0.603	0.591	0.595	0.563	0.464	0.664	0.582	0.594	0.680
DP@20	0.758	0.802	0.833	0.864	0.855	0.818	0.714	0.907	0.828	0.874	0.935

HS Trackers	MHT [16]	BAE-Net [27]	SiamBAG [18]	SSTtrack [22]	SENSE [28]	MMF-Net [13]	DaSSP-Net [24]	SpectralTrack [61]	HotMoE [44]	CCTrack [33]	SPIRIT [29]	PHTrack [23]	**HOTMM**
AUC	0.587	0.607	0.632	0.711	0.689	0.689	0.680	**0.725**	0.704	0.683	0.678	0.660	**0.749**
DP@20	0.880	0.877	0.892	**0.955**	0.950	0.931	0.916	0.953	0.935	0.923	0.924	0.918	**0.965**

highest competing DP@20, by 1.0%, confirming its superior tracking capability.

E. Comparison on HOT2023 Dataset

To evaluate the robustness of HOTMM across different spectral settings and diverse HS domains, we compare it with seven RGB trackers and nine HS trackers on the HOT2023 benchmark. The quantitative results are reported in Table II, while the corresponding precision and success plots for VIS2023, NIR2023, and RedNIR2023 are shown in Fig. 7(a)–(f).

1) VIS2023: As shown in the first two columns of Table II, HOTMM achieves the best performance, with an AUC of 0.685 and a DP@20 of 0.908. Compared with the second-ranked method, SSTtrack, HOTMM improves AUC by 2.8% and DP@20 by 3.6%. The corresponding curves in Fig. 7(a) and (b) further confirm its superior tracking performance.

2) NIR2023: As reported in the third and fourth columns of Table II, HOTMM also ranks first on NIR2023, achieving an AUC of 0.754 and a DP@20 of 0.965. Compared with DaSSP-Net, the runner-up method, HOTMM achieves gains of 0.8% in AUC and 1.3% in DP@20. This advantage is also reflected in the corresponding curves in Fig. 7(c) and (d).

3) RedNIR2023: As shown in the last two columns of Table II, HOTMM achieves an AUC of 0.590 and a DP@20 of 0.751 on RedNIR2023, slightly outperforming DaSSP-Net by 0.3% in AUC and 2.0% in DP@20. Although the curves of the two trackers intersect in Fig. 7(e) and (f), HOTMM remains more competitive under stricter evaluation thresholds. Specifically, HOTMM yields higher precision when the localization error threshold is below 30 pixels, and achieves better success when the IoU threshold exceeds 0.3. These results indicate that HOTMM provides more accurate target localization in this challenging HS domain.

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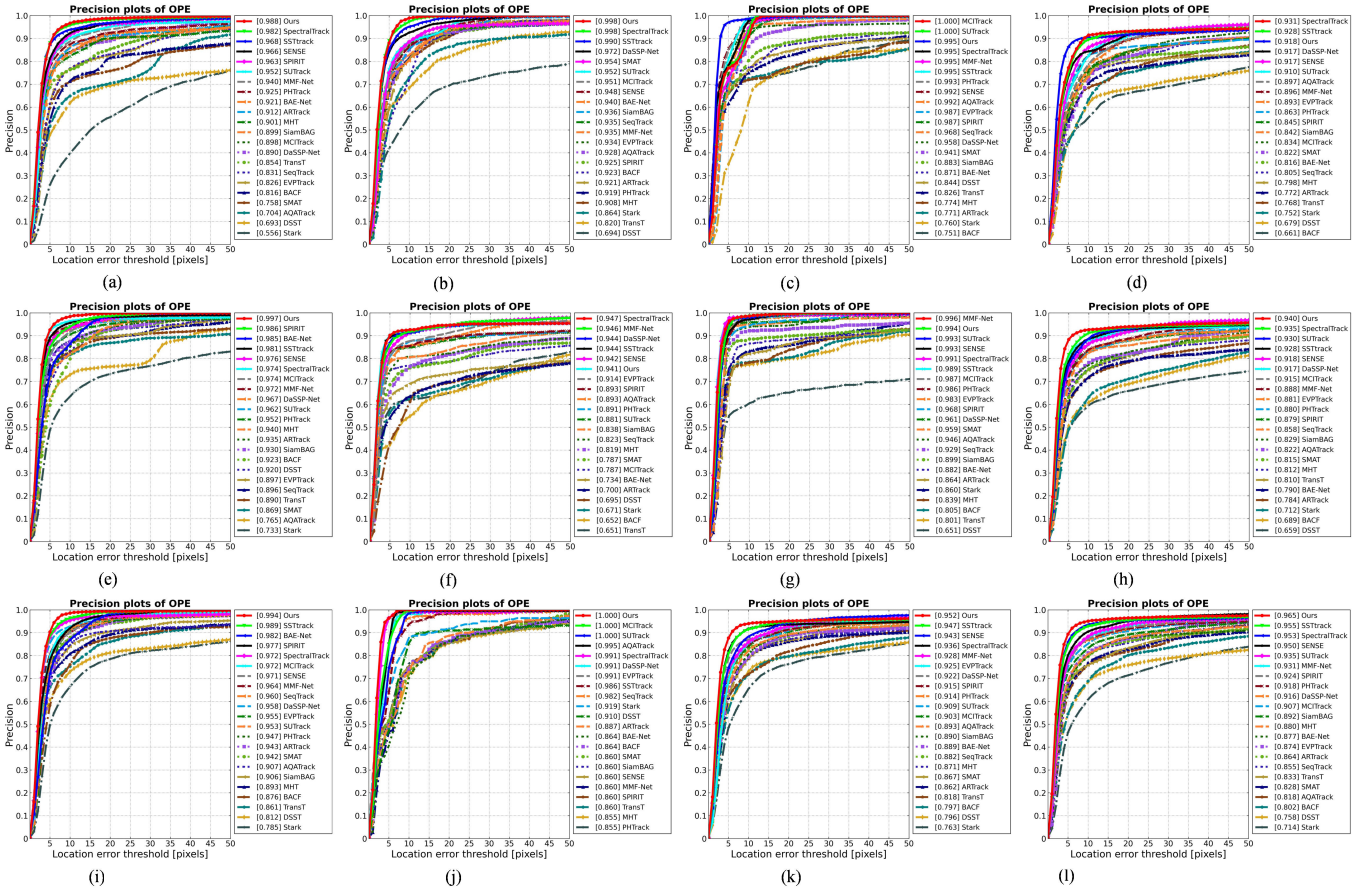


Fig. 6. Attribute-wise and overall precision plots of RGB and HS trackers on the HOT2020 dataset. (a)–(k) Results under the attributes BC, DEF, FM, IV, IPR, LR, MB, OCC, OPR, OV, and SV. (l) Overall precision plot.

F. Comparison on HOT2024 Dataset

To assess the generalization of HOTMM on a larger benchmark, we conduct additional comparisons on HOT2024. The quantitative results are reported in Table III, while the corresponding precision and success plots for VIS2024, NIR2024, and RedNIR2024 are shown in Fig. 7(g)–(l).

1) VIS2024: As shown in the first two columns of Table III, HOTMM achieves the best performance, with an AUC of 0.597 and a DP@20 of 0.776. Compared with the second-best AUC achieved by MCITrack, HOTMM improves AUC by 2.9%. For DP@20, HOTMM exceeds VIPT, which attains the second-best result on this metric, by 6.0%. The corresponding curves in Fig. 7(g) and (h) further support this result.

2) NIR2024: As reported in the middle two columns of Table III, HOTMM also ranks first on NIR2024, achieving an AUC of 0.755 and a DP@20 of 0.937. Compared with the strongest competing methods, HOTMM improves AUC by 3.6% over UBSTrack and DP@20 by 4.9% over SSTtrack. This advantage is also reflected in Fig. 7(i) and (j).

3) RedNIR2024: As shown in the last two columns of Table III, HOTMM achieves an AUC of 0.525 and a DP@20 of 0.681 on RedNIR2024, outperforming SSTtrack, the second-ranked method on both metrics, by 1.1% and 2.2%, respectively. The corresponding curves in Fig. 7(k) and (l) further confirm its advantage in this domain.

TABLE II
COMPARISON RESULTS ON HOT2023 DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE

Trackers	VIS2023		NIR2023		RedNIR2023	
	AUC	DP@20	AUC	DP@20	AUC	DP@20
SeqTrack [55]	0.569	0.805	0.639	0.872	0.426	0.540
AQATrack [56]	0.537	0.763	0.602	0.815	0.446	0.573
Stark [57]	0.431	0.640	0.433	0.628	0.374	0.499
MCITrack [32]	0.610	0.823	0.589	0.767	0.428	0.567
SMAT [58]	0.524	0.728	0.667	0.867	0.355	0.459
EVPTrack [59]	0.563	0.809	0.653	0.866	0.483	0.629
SUTrack [60]	0.612	0.840	0.681	0.883	0.444	0.579
BAE-Net [27]	0.571	0.824	0.451	0.786	0.416	0.557
Trans-DAT [20]	0.512	0.721	0.681	0.900	0.537	0.715
SENSE [28]	0.608	0.826	0.547	0.771	0.395	0.500
MMF-Net [13]	0.613	0.815	0.664	0.910	0.325	0.410
SEE-Net [36]	0.594	0.813	0.471	0.785	0.361	0.491
SSTtrack [22]	**0.657**	**0.872**	0.657	0.854	0.407	0.499
DaSSP-Net [24]	0.653	0.865	**0.746**	**0.952**	**0.587**	**0.731**
SPIRIT [29]	0.609	0.820	0.623	0.838	0.391	0.501
PHTrack [23]	0.581	0.795	0.536	0.780	0.417	0.515
HOTMM	**0.685**	**0.908**	**0.754**	**0.965**	**0.590**	**0.751**

The dashed line separates RGB and HS trackers.

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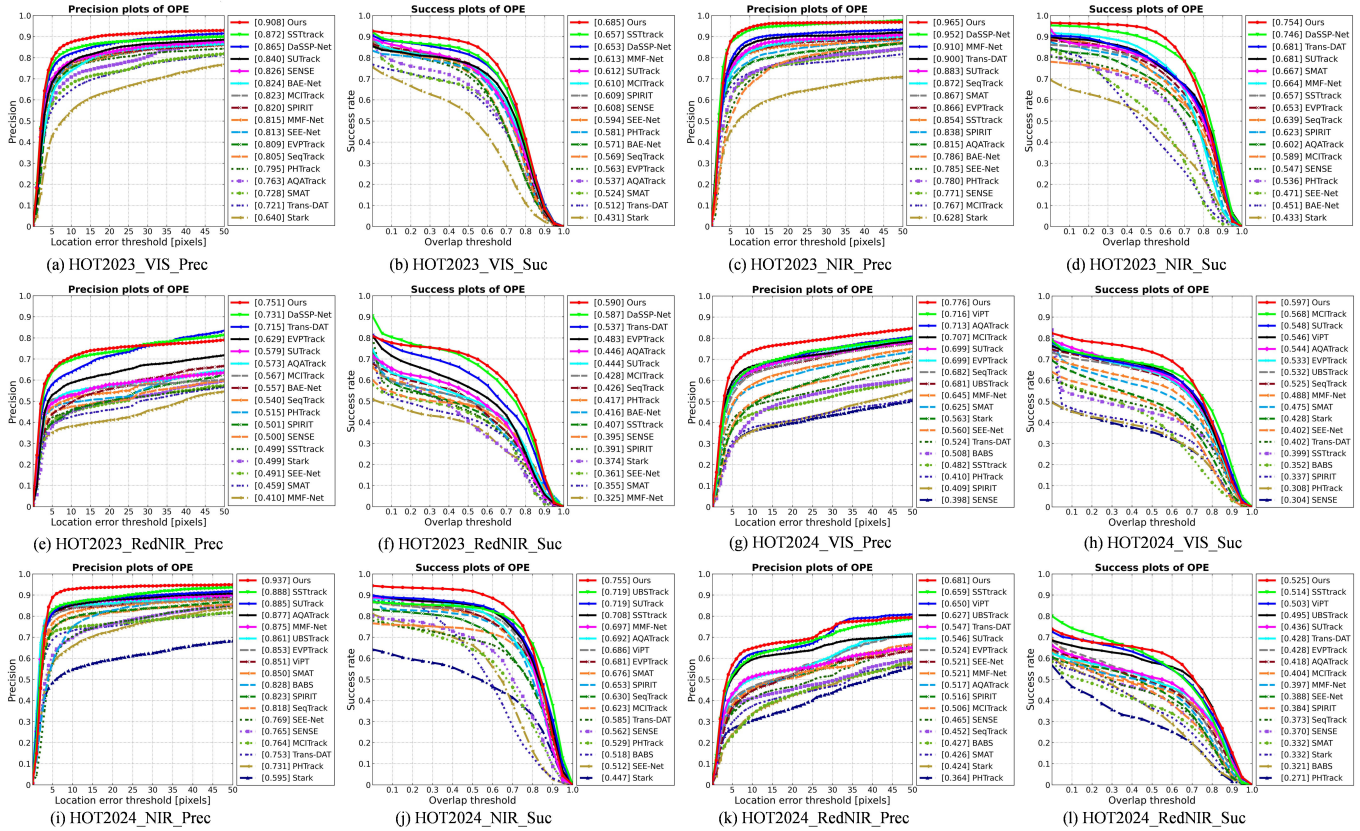


Fig. 7. Performance evaluation with RGB and HS trackers on the HOT2023 and HOT2024 datasets. (a)–(f) Precision and success plots for VIS2023, NIR2023, and RedNIR2023. (g)–(l) Corresponding results on VIS2024, NIR2024, and RedNIR2024, respectively.

G. Comparison on IMEC25 Dataset

In this experiment, we evaluate the performance of HOTMM in scenes dominated by numerous small targets by conducting quantitative comparisons on the IMEC25 dataset against 21 SOTA trackers. As reported in Table IV and Fig. 8, HOTMM achieves a DP@20 of 0.978 and an AUC of 0.740, ranking first overall among all compared methods. Specifically, HOTMM improves DP@20 by 1.4% over COALT, which ranks second on this metric, and exceeds PHTrack, the second-best performer in terms of AUC, by 3.0%, demonstrating robust tracking performance in such challenging scenarios. The inferior performance of the Siamese-based trackers listed in Table IV can be attributed to their limited capability in learning discriminative spectral representations, which restricts their effectiveness in tracking fine-scale targets. Even when compared with the strongest method in this category, SiamF, HOTMM delivers substantial gains of 4.6% in DP@20 and 8.3% in AUC, further confirming its superior tracking performance under small-target–dominated conditions.

H. Ablation Study

In this section, we conduct comprehensive ablation studies on the HOT2020 dataset to evaluate the contributions of the proposed components and design choices.

1) Contribution of Individual Components: This experiment evaluates the contribution of each proposed component, and the quantitative results are reported in Table V. The first row corresponds to the baseline adapted from MambaLCT [49], where an additional 3-D convolutional branch is introduced to incorporate HS information into the original framework. The remaining variants are constructed by

TABLE III

COMPARISON RESULTS ON HOT2024 DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE

Trackers	VIS2024		NIR2024		RedNIR2024	
	AUC	DP@20	AUC	DP@20	AUC	DP@20
SeqTrack [55]	0.525	0.682	0.630	0.818	0.373	0.452
AQATrack [56]	0.544	0.713	0.692	0.877	0.418	0.517
Stark [57]	0.428	0.563	0.447	0.595	0.332	0.424
VIPT [62]	0.546	**0.716**	0.686	0.851	0.503	0.650
MCITrack [32]	**0.568**	0.707	0.623	0.764	0.404	0.506
SMAT [58]	0.475	0.625	0.676	0.850	0.332	0.426
EVPTrack [59]	0.533	0.699	0.681	0.853	0.428	0.524
SUTrack [60]	0.548	0.699	0.717	0.885	0.436	0.546
BABS [35]	0.352	0.508	0.518	0.828	0.321	0.427
Trans-DAT [20]	0.402	0.524	0.585	0.753	0.428	0.547
SENSE [28]	0.304	0.398	0.562	0.765	0.370	0.465
MMF-Net [13]	0.488	0.645	0.697	0.875	0.397	0.521
SSTtrack [22]	0.399	0.482	0.708	**0.888**	**0.514**	**0.659**
SEE-Net [36]	0.402	0.560	0.512	0.769	0.388	0.521
UBSTrack [25]	0.532	0.681	**0.719**	0.861	0.495	0.627
SPIRIT [29]	0.337	0.409	0.653	0.823	0.384	0.516
PHTrack [23]	0.308	0.410	0.529	0.731	0.271	0.364
HOTMM	**0.597**	**0.776**	**0.755**	**0.937**	**0.525**	**0.681**

The dashed line separates RGB and HS trackers.

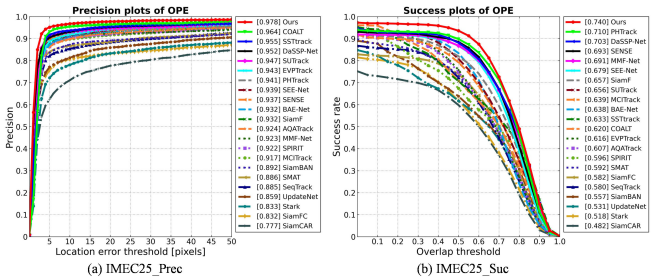
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TABLE IV

COMPARISON WITH SOTA TRACKERS ON THE IMEC25 DATASET. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN RED AND BLUE

RGB Trackers	SiamFC [10]	SiamBAN [11]	SiamCAR [63]	UpdateNet [64]	SeqTrack [55]	AQATrack [56]	Stark [57]	MCITrack [32]	SMAT [58]	EVPTrack [59]	SUTrack [60]
AUC	0.582	0.557	0.482	0.531	0.580	0.607	0.518	0.639	0.592	0.616	0.656
DP@20	0.832	0.892	0.777	0.859	0.885	0.924	0.833	0.917	0.886	0.943	0.947

HS Trackers	SiamF [12]	BAE-Net [27]	SEE-Net [36]	SSTtrack [22]	SENSE [28]	MMF-Net [13]	DaSSP-Net [24]	COALT [21]	SPIRIT [29]	PHTrack [23]	**HOTMM**
AUC	0.657	0.638	0.679	0.633	0.693	0.691	0.703	0.620	0.596	**0.710**	**0.740**
DP@20	0.932	0.932	0.939	0.955	0.937	0.923	0.952	**0.964**	0.922	0.941	**0.978**



(a) IMEC25_Prec (b) IMEC25_Suc

Fig. 8. Performance evaluation with RGB and HS trackers on the IMEC25 dataset. (a) Precision plot. (b) Success plot.

TABLE V

ABLATION STUDY OF HOTMM ON HOT2020 DATASET

HDA	MDME	TMM	AUC	ΔAUC	DP@20	ΔDP@20
✗	✗	✗	0.665	-	0.909	-
✓	✗	✗	0.674	+0.9%	0.916	+0.7%
✗	✓	✗	0.710	+4.5%	0.941	+3.2%
✗	✗	✓	0.689	+2.4%	0.928	+1.9%
✓	✓	✗	0.714	+4.9%	0.946	+3.7%
✓	✗	✓	0.691	+2.6%	0.932	+2.3%
✗	✓	✓	0.741	+7.6%	0.962	+5.3%
✓	✓	✓	**0.749**	**+8.4%**	**0.965**	**+5.6%**

The best results are highlighted in red.

incrementally adding the proposed modules to this baseline. As shown in the second row, introducing HDA improves AUC and DP@20 by 0.9% and 0.7%, indicating the benefit of lightweight cross-domain adaptation. Adding MDME in the third row yields larger gains of 4.5% in AUC and 3.2% in DP@20, demonstrating the effectiveness of the proposed adaptive spatial–spectral modeling. In the fourth row, TMM improves AUC and DP@20 by 2.4% and 1.9%, confirming the value of temporal memory modeling. Rows five to seven further evaluate pairwise combinations of the proposed modules. Among them, combining MDME with TMM produces the largest gains, improving AUC and DP@20 by 7.6% and 5.3%, which suggests that the two modules are complementary. Finally, adding HDA to this setting achieves the best performance, raising the gains to 8.4% in AUC and 5.6 % in DP@20.

2) Contribution of the False-Color Branch: To examine the contribution of the false-color branch, we compare the full dual-branch setting with a variant that removes the false-color input, and the results are reported in Table VI. The complete model with both HS and false-color branches achieves the best performance, reaching an AUC of 0.749 and a DP@20 of 0.965. After removing the false-color branch, the performance drops to 0.737 in AUC and 0.950 in DP@20, corresponding to declines of 1.2% and 1.5%. These results indicate that the false-color branch provides complementary structural cues beyond the HS representation alone. In particular, it offers a useful structural prior that facilitates spatial modeling and improves target localization, while the HS branch remains the primary carrier of discriminative spectral information. Therefore, the dual-branch design is justified, as the false-color branch serves as an effective auxiliary component rather than a redundant addition.

TABLE VI

ABLATION STUDY OF THE FALSE-COLOR BRANCH ON HOT2020 DATASET

Method	AUC	ΔAUC	DP@20	ΔDP@20
HS + False-color Branches	**0.749**	-	**0.965**	-
w/o False-color Branch	0.737	-1.2%	0.950	-1.5%

The best results are highlighted in red.

TABLE VII

ABLATION STUDY OF MDME MODULE ON HOT2020 DATASET

Method	AUC	ΔAUC	DP@20	ΔDP@20
MDME	**0.749**	-	**0.965**	-
w/o SRME	0.715	-3.4%	0.941	-2.2%
w/o SSME	0.723	-2.6%	0.948	-1.7%
w/o SRME&SSME	0.691	-5.8%	0.932	-3.3%

The best results are highlighted in red.

3) Internal Design of MDME: To examine the contribution of the MDME design, we evaluate different component configurations by selectively removing SRME and SSME. As reported in Table VII, removing either component degrades both AUC and DP@20, indicating that spatial and spectral modeling are both important for MDME. Specifically, compared with the complete MDME, removing SRME or SSME causes AUC drops of 3.4% and 2.6%, and DP@20 drops of 2.2% and 1.7%, respectively, while removing both components leads to larger declines of 5.8% in AUC and 3.3% in DP@20. These results demonstrate the complementary roles of SRME and SSME in spatial–spectral representation learning.

4) Number of Experts Selected in SRME: This experiment evaluates the influence of the top-k selection strategy in SRME, and the results are reported in Table VIII. Increasing k from 1 to 2 improves AUC and DP@20 by 1.8% and 1.2%, suggesting that combining two directional experts provides complementary spatial cues beyond single-direction modeling. However, further increasing k leads to performance

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TABLE VIII
ABLATION STUDY OF THE TOP-k NUMBERS IN SRME

Value of k	AUC	ΔAUC	DP@20	ΔDP@20
k=1	0.731	-	0.953	-
k=2	**0.749**	**+1.8%**	**0.965**	**+1.2%**
k=3	0.744	+1.3%	0.962	+0.9%
k=4	0.732	+0.1%	0.955	+0.2%

The best results are highlighted in red.

TABLE IX
ABLATION STUDY OF SCANNING STRATEGY IN SSME

Method	AUC	ΔAUC	DP@20	ΔDP@20
w/o SSME	0.723	-	0.948	-
Forward Scan	0.744	+2.1%	0.962	+1.4%
Backward Scan	0.741	+1.8%	0.961	+1.3%
Bidirectional Scan	**0.749**	**+2.6%**	**0.965**	**+1.7%**

The best results are highlighted in red.

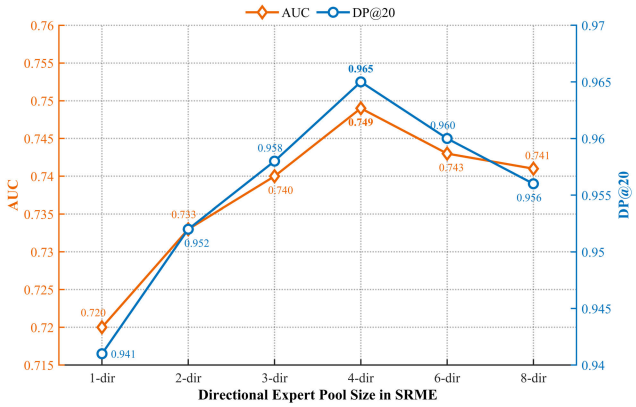


Fig. 9. Effect of the directional expert pool size on tracking performance.

degradation. When $k = 3$, the additional experts bring limited benefit, indicating that excessive directional integration introduces redundant spatial information. When $k = 4$, all directional experts participate in the fusion, causing SRME to behave more like a dense aggregation scheme. As a result, AUC and DP@20 drop by 1.7% and 1.0%, compared with the best setting. Overall, $k = 2$ provides the best balance between spatial diversity and feature selectivity, confirming the effectiveness of sparse directional routing in SRME.

5) Directional Expert Pool Size in SRME: We further evaluate the influence of the directional expert pool size in SRME, and the results are shown in Fig. 9. Here, "dir" on the horizontal axis denotes the number of directional scan candidates. We compare representative scan configurations with clear structural interpretations rather than exhaustively enumerating all possible scan combinations. Specifically, 1-dir uses a single row-major scan, while 2-dir adopts bidirectional row-major scans. Based on this, 3-dir introduces one additional column-major scan as an intermediate asymmetric configuration, and 4-dir corresponds to our default design with bidirectional row-major and column-major scans. To examine whether more directional candidates are beneficial, 6-dir further adds bidirectional row-wise serpentine scans to 4-dir, while 8-dir additionally incorporates bidirectional column-wise serpentine scans. For a fair comparison, k is set to 2 for all configurations with more than one candidate, following the best setting in the top-k ablation.

As shown in Fig. 9, enlarging the expert pool from 1-dir to 4-dir leads to consistent performance gains. This trend indicates that progressively enriching the directional scan set expands the spatial propagation paths, enabling SRME to capture more complete 2-D contextual information. The 1-dir baseline achieves an AUC of 0.720 and a DP@20 of 0.941. With the reverse row-major scan introduced, 2-dir improves AUC and DP@20 by 1.3% and 1.1%, indicating the benefit of bidirectional propagation. After adding one column-major scan, 3-dir brings additional gains of 0.7% in AUC and 0.6% in DP@20 over 2-dir, suggesting that complementary traversal orientations help enhance 2-D spatial dependency modeling. The proposed 4-dir setting achieves the best performance, yielding gains of 0.9% in AUC and 0.7% in DP@20 over 3-dir, confirming the effectiveness of the complete bidirectional row-major and column-major scan set. However, further enlarging the expert pool does not yield continuous improvements. Compared with 4-dir, 6-dir decreases AUC and DP@20 by 0.6% and 0.5%, while 8-dir further drops by 0.8% and 0.9%. This suggests that additional directional experts provide limited complementary information beyond the compact four-directional set. Under the sparse top-k routing strategy, a larger candidate pool also increases the routing space and may reduce the training frequency of individual experts. Therefore, the 4-dir design provides a favorable tradeoff between directional diversity, routing stability, and tracking accuracy.

6) Scanning Strategy of SSME: In this section, we analyze the scanning strategy of SSME by evaluating three configurations: forward spectral scanning, backward spectral scanning, and bidirectional spectral scanning. The baseline corresponds to the model without SSME. As summarized in Table IX, the bidirectional scanning strategy consistently outperforms both single-direction counterparts, achieving the highest tracking accuracy. Specifically, relative to the baseline, bidirectional spectral scanning improves AUC and DP@20 by 2.6% and 1.7%. This advantage stems from the noncausal nature of spectral dependencies in HS data. While single-direction scanning aggregates contextual information predominantly from one side of the spectrum, bidirectional scanning enables each spectral token to integrate complementary cues from both lower and higher bands. Moreover, the symmetric fusion of forward and backward scans with shared parameters provides a more balanced aggregation of spectral context, leading to more stable and discriminative spectral representations.

7) Number of SSME Layers: Since spectral information in HS data mainly provides discriminative cues for target identification, we further study SSME depth by varying the number of SSME layers. As shown in Table X, the best performance is achieved when two SSME layers are used, whereas both shallower and deeper configurations lead to lower AUC and DP@20. Using fewer layers limits the ability to capture sufficient spectral dependencies, while stacking more layers introduces redundant spectral aggregation and over-smoothing across bands, which weakens feature discriminability. Based

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TABLE X
ABLATION STUDY OF THE NUMBER OF SSME

Number	AUC	ΔAUC	DP@20	ΔDP@20
0	0.723	-	0.948	-
1	0.740	+1.7%	0.961	+1.3%
2	**0.749**	**+2.6%**	**0.965**	**+1.7%**
3	0.742	+1.9%	0.957	+0.9%
4	0.727	+0.4%	0.952	+0.5%

The best results are highlighted in red.

TABLE XI
ABLATION STUDY OF TEMPORAL MEMORY DESIGN

Method	AUC	ΔAUC	DP@20	ΔDP@20
TMM	**0.749**	-	**0.965**	-
Simple Temporal Memory	0.728	-2.1%	0.950	-1.5%

The best results are highlighted in red.

on this observation, we adopt a lightweight design in which SSME is activated only in the first two layers together with SRME to generate spatial–spectral prompts, while the subsequent layers retain only SRME. This design reduces complexity while achieving the best tracking performance.

8) Comparison With Simple Temporal Memory: To verify that the gain of TMM comes from its temporal design rather than merely from introducing historical information, we replace TMM with a simple temporal memory module and report the results in Table XI. For fairness, the rest of the architecture remains unchanged, and only the temporal update mechanism is replaced. Specifically, the frame-level descriptor extracted by MGTP is retained, while the Mamba-based state-space recurrence is replaced by a simple recursive update using an exponential moving average of the current descriptor and the accumulated history. As shown in Table XI, this replacement reduces AUC from 0.749 to 0.728 and DP@20 from 0.965 to 0.950, corresponding to drops of 2.1% and 1.5%. These results show that the gain is not due solely to temporal memory, but to the stronger long-range temporal modeling of TMM.

9) Hidden State Dimension: Since both MDME and TMM employ Mamba-based state space modeling, we further study the effect of the hidden state dimension, which controls the capacity of state-space memory. As shown in Fig. 10, the best tracking accuracy is achieved when the hidden state dimension is set to 16. When the dimension is smaller, the limited memory capacity restricts the modeling of informative dependencies. In contrast, using a larger hidden dimension introduces redundant states, which degrades representation efficiency and hurts performance. Therefore, a hidden state dimension of 16 is adopted as it provides the best balance between modeling capacity and stability.

I. Model Complexity and Inference Efficiency

To assess the practicality of HOTMM, we further compare its model size, FLOPs, and inference speed with representative SOTA HS trackers on the HOT2020 dataset. As reported in Table XII, HOTMM contains 76.7-M total parameters,

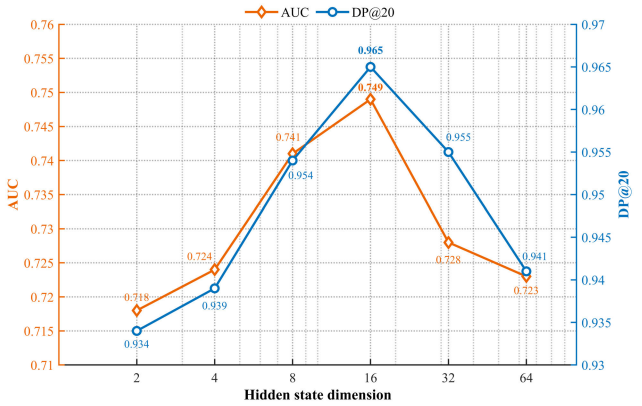


Fig. 10. Effect of varying hidden state dimension on tracking performance.

TABLE XII
EFFICIENCY EVALUATION IN TERMS OF MODEL PARAMETERS, FLOPS, AND INFERENCE SPEED

Trackers	Trainable Params (M)	Total Params (M)	FLOPs (G)	Speed (FPS)
PHTrack [23]	0.2	51.6	604.7	15.2
SSTtrack [22]	0.6	72.6	125.0	15.5
DASSP-Net [24]	2.2	94.7	46.3	27.5
SpectralTrack [61]	3.9	75.9	75.6	36.7
SEE-Net [36]	7.5	53.9	244.2	8.7
MMF-Net [13]	8.3	45.1	352.6	6.1
SiamF [12]	10.3	56.7	227.7	3.0
HOTMM	**4.4**	**76.7**	**123.8**	**26.4**

The bold row denotes the results of our method.

among which 4.4 M are trainable during HOT adaptation. This indicates that only a small proportion of parameters are updated, consistent with the parameter-efficient adaptation strategy adopted in HOTMM. In terms of computational cost, HOTMM requires 123.8 GFLOPs, comparable to SSTtrack and lower than PHTrack, SEE-Net, MMF-Net, and SiamF. For inference speed, HOTMM achieves 26.4 FPS, outperforming PHTrack, SSTtrack, SEE-Net, MMF-Net, and SiamF, while being slower than DaSSP-Net and SpectralTrack. Although HOTMM is not the lightest tracker, it maintains practical inference efficiency while achieving stronger tracking performance. These results indicate that the proposed MDME and TMM introduce manageable computational overhead and provide a favorable tradeoff between tracking accuracy and efficiency.

J. Attributes-Based Evaluation

The performance of HOTMM is evaluated on 11 representative challenging tracking attributes in comparison with SOTA trackers, as shown in Figs. 5 and 6. The corresponding precision and success scores for each attribute are reported in Table XIII. HOTMM achieves the best AUC on ten attributes and the highest DP@20 on seven attributes. In particular, it ranks first on both metrics under the BC, DEF, IPR, OCC, OPR, OV, and SV attributes, indicating its strong robustness across diverse tracking conditions. Moreover, relative to the second-best tracker, HOTMM improves the AUC by 3.3%,

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TABLE XIII

ATTRIBUTE-WISE DP@20 AND AUC SCORES ON THE HOT2020 DATASET. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE INDICATED IN RED, BLUE, AND GREEN

Trackers	BC	DEF	FM	IV	IPR	LR	MB	OCC	OPR	OV	SV
DSST [51]	0.693/0.441	0.694/0.444	0.844/0.582	0.679/0.385	0.920/0.639	0.695/0.355	0.651/0.431	0.659/0.437	0.812/0.559	0.910/0.573	0.796/0.495
BACF [52]	0.816/0.526	0.923/0.628	0.751/0.562	0.661/0.400	0.923/0.650	0.652/0.385	0.805/0.595	0.689/0.478	0.876/0.622	0.864/0.581	0.797/0.540
TransT [53]	0.854/0.616	0.820/0.639	0.826/0.561	0.768/0.513	0.890/0.692	0.651/0.437	0.801/0.582	0.810/0.590	0.861/0.662	0.860/0.622	0.818/0.591
ARTrack [54]	0.912/0.650	0.921/0.674	0.771/0.557	0.772/0.465	0.935/0.688	0.700/0.404	0.864/0.567	0.784/0.535	0.943/0.694	0.887/0.540	0.862/0.579
SeqTrack [55]	0.831/0.569	0.935/0.692	0.968/0.640	0.805/0.553	0.896/0.665	0.823/0.529	0.929/0.612	0.858/0.595	0.960/0.709	0.982/0.751	0.882/0.621
AQATrack [56]	0.704/0.490	0.928/0.675	0.992/0.624	0.897/0.596	0.765/0.569	0.893/0.573	0.946/0.621	0.822/0.584	0.907/0.666	0.995/0.790	0.893/0.621
Stark [57]	0.556/0.346	0.864/0.611	0.760/0.425	0.752/0.480	0.733/0.503	0.671/0.408	0.860/0.546	0.712/0.473	0.785/0.545	0.919/0.679	0.763/0.509
MCITrack [32]	0.898/0.664	0.951/0.730	1.000/0.703	0.834/0.586	0.974/0.747	0.787/0.558	0.987/0.705	0.915/0.666	0.972/0.757	1.000/0.791	0.903/0.666
SMAT [58]	0.758/0.551	0.954/0.729	0.941/0.625	0.822/0.515	0.869/0.665	0.787/0.508	0.959/0.689	0.815/0.574	0.942/0.713	0.860/0.652	0.867/0.600
EVPTrack [59]	0.826/0.564	0.934/0.677	0.987/0.624	0.893/0.585	0.897/0.651	0.914/0.567	0.983/0.637	0.881/0.608	0.955/0.697	0.991/0.809	0.925/0.636
SUTrack [60]	0.952/0.698	0.952/0.742	1.000/0.700	0.910/0.631	0.962/0.739	0.881/0.618	0.993/0.698	0.930/0.679	0.953/0.747	1.000/0.795	0.909/0.666
MHT [16]	0.901/0.607	0.908/0.664	0.774/0.541	0.798/0.481	0.940/0.670	0.819/0.477	0.839/0.560	0.812/0.565	0.893/0.644	0.855/0.627	0.871/0.576
BAE-Net [27]	0.921/0.650	0.940/0.679	0.871/0.607	0.816/0.528	0.985/0.699	0.734/0.491	0.882/0.594	0.790/0.555	0.982/0.701	0.864/0.523	0.889/0.606
SiamBAG [11]	0.899/0.646	0.936/0.692	0.883/0.614	0.842/0.535	0.930/0.702	0.838/0.581	0.899/0.648	0.829/0.597	0.906/0.682	0.860/0.637	0.890/0.623
SSTtrack [22]	0.968/0.731	0.990/0.778	0.995/0.733	0.928/0.653	0.981/0.770	0.944/0.662	0.989/0.742	0.928/0.672	0.989/0.790	0.986/0.768	0.947/0.706
SENSE [28]	0.966/0.720	0.948/0.708	0.992/0.744	0.917/0.611	0.976/0.749	0.942/0.622	0.993/0.728	0.918/0.645	0.971/0.735	0.860/0.632	0.943/0.673
MMF-Net [13]	0.940/0.707	0.935/0.722	0.995/0.704	0.896/0.593	0.972/0.778	0.946/0.626	0.996/0.761	0.888/0.649	0.964/0.765	0.860/0.607	0.928/0.679
DaSSP-Net [24]	0.890/0.655	0.972/0.761	0.958/0.707	0.917/0.657	0.967/0.759	0.944/0.654	0.961/0.721	0.917/0.681	0.958/0.760	0.991/0.808	0.922/0.687
SpectralTrack [61]	0.982/0.749	0.998/0.791	0.995/0.740	0.931/0.682	0.974/0.781	0.947/0.669	0.991/0.756	0.935/0.701	0.972/0.783	0.991/0.832	0.936/0.714
SPIRIT [29]	0.963/0.713	0.925/0.722	0.987/0.714	0.845/0.569	0.986/0.758	0.893/0.630	0.968/0.722	0.879/0.642	0.977/0.749	0.860/0.690	0.915/0.663
PHTrack [23]	0.925/0.678	0.919/0.674	0.993/0.734	0.863/0.580	0.952/0.723	0.891/0.584	0.986/0.729	0.880/0.619	0.947/0.708	0.855/0.610	0.914/0.647
HOTMM (Ours)	0.988/0.756	0.998/0.802	0.995/0.777	0.918/0.708	0.997/0.809	0.941/0.674	0.994/0.747	0.940/0.729	0.994/0.811	1.000/0.850	0.952/0.749

The dashed line separates RGB and HS trackers.

2.8%, and 3.5% on the FM, IPR, and SV attributes, respectively. These results suggest that HOTMM maintains reliable target discrimination in complex HS scenes while effectively handling attribute-specific tracking challenges.

K. Visualizations of Tracking Results

Representative visual results are shown in Fig. 4 to qualitatively evaluate the tracking performance of HOTMM. As illustrated, HOTMM maintains accurate target localization under various challenging conditions. For example, in the "car1" sequence with IV, OCC, DEF, and BC attributes, HOTMM successfully tracks the small bicycle across frames with well-aligned bounding boxes and accurate scale estimation, whereas other trackers either drift from the target (e.g., SPIRIT and PHTrack) or produce inaccurate scales (e.g., SEENet and AQATrack). In more challenging sequences, such as "doublecar7" with IV, OCC, DEF, BC, and LR attributes, and "doublebicycle" with SV, OCC, DEF, and BC attributes, all competing trackers fail to maintain reliable tracking, while HOTMM consistently localizes the target throughout the sequence. These results further demonstrate the robustness of HOTMM in complex HS tracking scenarios.

V. CONCLUSION

In this article, we presented HOTMM, a novel Mamba-based HOT framework that combines direction-aware spatial–spectral prompting with memory-guided temporal modeling. MDME introduces a routed-and-shared MoE design to adaptively capture directional spatial structures and aggregate group-wise spectral features, producing content-adaptive prompts for enhancing RGB-pretrained tracking representations. TMM maintains a long-range temporal memory to encode persistent target evolution and refine target descriptors over time. Extensive experiments on eight HOT benchmarks demonstrate the superiority of HOTMM over SOTA trackers. Future work will further investigate stronger cross-domain generalization and more self-contained HS modeling. In addition, adaptive spectral organization can be explored to complement the proposed group-wise spectral aggregation, enabling spectral dependencies to be modeled more flexibly under different tracking scenarios.

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